

Event Message Reference Manual

BASE24[®]



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Preface

ACI provides BASE24 event message documentation on HP NonStop subvolumes for you to print and use as you require. The ***BASE24 Event Message Reference Manual*** provides you with instructions on how to locate, access, and print this event message documentation.

In addition, this manual provides an overview of event messages and how they are generated by BASE24 processes. It describes the logging facilities available with BASE24 products and the event message formats used by BASE24 products. It also describes the Manual Generator utility, which can be used with BASE24 products to automatically generate event message and library documentation files.

Audience

The ***BASE24 Event Message Reference Manual*** can be used by several different audiences. It is designed for use by anyone seeking a basic understanding of BASE24 event messages. It is also intended for anyone interested in learning how to access, print, and use the BASE24 event message documentation, or for anyone who needs to generate event message or library documentation files for object-oriented products.

Prerequisites

Persons using this manual should have a working knowledge of a HP NonStop editor and other associated HP NonStop tools such as TGAL in order to access and print event message documentation and TACL to run the Manual Generator utility.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Base Files Maintenance Manual*** describes of the BASE24 standard function keys.
- The ***BASE24-pos APACS 30/40 Device Handler User's Guide*** documents event messages for the APACS 30/40 POS Device Handler process.
- The ***BASE24 EPS-Net Interchange Interface Manual*** documents event messages for the EPSNet Interchange Interface process.
- The ***BASE24 LINK ISO Interchange Interface Manual*** documents event messages for the LINK Interchange Interface process.

This manual also contains references to the ***NET24-XPNET Event Management Manual*** for detailed information about the Event Management Service (EMS).

In addition to the BASE24 and NET24-XPNET publications, the ***BASE24 Event Message Reference Manual*** contains references to the informal ACI ***Object-Oriented Programming Standards Manual*** for information on using updated EMS templates and TAL libraries. The ***BASE24 Event Message Reference Manual*** also references the following HP NonStop publications:

- The ***EMS Manual*** provides detailed information on the HP NonStop Event Management Service (EMS).
- The ***Guardian User's Guide*** contains information on mapping file defines, which may be required prior to running the Manual Generator utility.
- The ***PS TEXT FORMAT Reference Manual*** provides information for using the HP NonStop PS Text Format (TFORM) product. The TFORM product can be used to print event message and TAL library documentation files.

- The *TACL Reference Manual* provides information on mapping file defines and running programs from a TACL prompt.
- The *TGAL Manual* provides instructions for using the HP NonStop TGAL product. The TGAL product can be used to print event message and TAL library documentation files.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” provides an overview of event messages and how they are generated by BASE24 network processes and BASE24 shared processes. It describes the logging facilities available with BASE24 products and the event message formats used by BASE24 products.

Section 2, “How Event Messages Are Documented,” describes how event message documentation is set up, file organization and format, event message organization and format, and the files where event messages are documented. This section includes examples of event message documentation.

Section 3, “Using Event Message Documentation,” describes the procedures you can use to locate and print event messages in the documentation. It includes the options for printing a single file and the options for printing multiple files at one time. It also includes the procedures you can use to access other files contained on the event message documentation subvolume.

Section 4, “The Manual Generator Utility,” describes the Manual Generator utility, which can be used to automatically generate event message and TAL library documentation for BASE24 products.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, BA-AE000-07 for the ***BASE24 Event Message Reference Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Dec-2008 for December 2008) indicates the issue of the manual. The software release information (for example, R6.0v9 for release 6.0, version 9) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in this Manual

This section explains how the security best practices symbol is used in this manual and how command syntax and field descriptions are documented in this manual. It also explains event message terminology.

Security Best Practices Symbol



This manual uses the symbol shown at the left to alert you to cross-industry best practices for protecting sensitive data that is stored, processed, or transmitted in a BASE24 system. Whenever you see this symbol, you should carefully review the information provided next to the symbol to ensure you are implementing cross-industry best practices for securing data. Refer to the ***BASE24 Security Best Practices*** for more information about configuring the BASE24 product to comply with cross-industry best practices.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
<i>lowercase italics</i>	Identify variables that you must provide.
Brackets []	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar ().

Notation	Meaning
Vertical bar	Separates alternatives in a list of items.
Ellipsis ...	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Field Descriptions

Each field appearing on a files maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from Log Message Configuration File (LMCF).

Example

PRODUCT MODULE ID — Specifies the product module ID (PMID) for the process for which message information is to be read, updated, or deleted. A list of valid product module IDs is included at the end of this section.

Field Length: 1–16 alphanumeric characters
Required Field: No
Default Value: No default value

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the files maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term <i>alphanumeric</i> includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term <i>hexadecimal</i> includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is <i>System Protected</i>.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the F8 key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

Event Message Terminology

The term *event message* is synonymous with the term *log message* used in other ACI manuals.

1: Introduction

This section provides an overview of event messages and how they are generated by BASE24 processes. It contains information to assist readers in understanding the basic concepts and components of event messages and defines the formats they see when event messages are generated. It also introduces the procedures used to automatically generate event message documentation for BASE24 products. This section includes discussions of the following topics:

- Event Message System (EMS) logging facility
- Identifying the originators of event messages
- PCI data security standards
- Log Message Configuration File (LMCF) and LMCF screen
- Event message format

BASE24 Event Messages

BASE24 event messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are the network operator's primary link to what is actually occurring in the system. BASE24 event messages provide the operator with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

The HP NonStop Event Management Service (EMS) is the logging facility used for all BASE24 products. The following paragraphs describe the approach used by EMS to categorize, format, and route event messages. For more information on EMS, refer to the *NET24-XPNET Event Management Manual* and the HP NonStop *EMS Manual*.

EMS Tokens

All BASE24 product event messages are sent to EMS in the form of tokens. Several tokens represent values that occur in all BASE24 messages. Some are header tokens provided by the EMS subsystem and others are provided by the XPNET subsystem. For information about the header tokens provided by the EMS subsystem, refer to the *NET24-XPNET Event Management Manual* and the HP NonStop *EMS Manual*. Tokens set by BASE24 products in the text of event messages are described in the online event message documentation for the events in which they are displayed.

Using the Event Management Service (EMS)

The HP NonStop Event Management Service (EMS) routes event messages based on filters, tokens, and multiple distributor processes. Event messages can be sent to terminals, printers, or files for display or storage.

When using EMS, the appearance of the event messages varies depending on parameters selected by the operator. Through use of the EMSperus utility, operators can select parameters that determine the information included in the event message, the order in which the information is displayed, and the format in which the information is displayed. In addition, through the use of filters, operators choose to display specific messages, while suppressing others. For example, an operator may want only event messages generated by the XPNET process to be displayed. In this case, a filter can be written to allow only XPNET process event messages to be displayed.

Customers can use the Log Message Configuration File (LMCF) to modify event messages. Refer to [“Modifying an LMCF Record”](#) and [“Using the MASK and MAX Directives”](#) later in this section for information on modifying event messages.

Identifying the Originator of an Event Message

Network operators must be able to recognize the product modules that generate various event messages in order to locate event messages in the documentation and resolve problems. The product module that generated a particular event message can be identified when EMS is used as the primary logging facility. When EMS is used, the product module ID (PMID) is included in the event message.

PMIDs uniquely identify processes, programs, or utilities within the BASE24 system that generate event messages. PMIDs are abbreviations of product module names that are hard coded in BASE24 products and are used as a way to identify event messages issued by a specific product module. PMIDs are included in each event message handled by EMS in the subsystem ID. Subsystem IDs for BASE24 products are displayed in the following format: *ACI.pmid.version*, where *pmid* is the product module ID and *version* is the current version of the product module. Both the PMID and version can be displayed in integer or text format. This manual documents the text format of PMIDs. The integer equivalent for each PMID is provided in the \$SYSTEM.SCRIBE.EMCLDDLS file.

ACI uses standard PMIDs to uniquely identify product modules within BASE24 products. All PMIDs have a maximum length of eight characters which makes them compatible with EMS. Standard BASE24 PMIDs identify the BASE24 product (e.g., BASE24-atm, BASE24-pos, BASE24-teller) and the module within the product that generated the event messages (e.g., Authorization, Settlement Initiator, Device Handler). The first two characters of the PMID represent the BASE24 product identifier. The following list contains the valid standard BASE24 product identifiers:

AT	=	BASE24-atm
BA	=	Base/Shared
CU	=	BASE24-check auth
FS	=	BASE24-frequent shopper
IS	=	Integrated Server (i.e., core, kernel, and integrated endpoint)
MA	=	BASE24-mail
TB	=	BASE24-telebanking
TP	=	BASE24-billpay
PS	=	BASE24-pos
RF	=	BASE24-refunds
SW	=	BASE24 interchanges
TR	=	BASE24-teller
XP	=	NET24-XPNET

The remaining characters represent the module within the product that generated the event messages. This can be up to six characters. For a list of BASE24 PMIDs and their associated process names, refer to section 2.

Note: There may be some exceptions to the naming conventions used for PMIDs. However, generally, PMIDs are named by product and module.

Multiple PMIDs

In some cases, multiple PMIDs can be associated with the same process. When this occurs, each PMID is used to identify a specific module within the process. Event messages for each of these different modules are documented in separate files. For example, a process such as the ISO Host Interface process is divided into product-specific modules, each identified by a different PMID. Each of these PMIDs may have a file residing on the event message subvolumes depending on the products the customer supports. Therefore, the ISO Host Interface process may require a file for BAHISO for its Base event messages, ATHISO for its BASE24-atm event messages, FHMHISO for its BASE24-from host maintenance event messages, PSHISO for its BASE24-pos event messages, TBHISO for its BASE24-telebanking event messages, and TLRHISO for its BASE24-teller event messages. For a listing of PMIDs, refer to section 2.

Generator ID

The generator ID is an 8-character string defined and passed by the process to identify the PPD name or symbolic name of the process for which the event message is being generated. For all XPNET processes, this token is set to the value XPNET. For some products, such as BASE24-billpay, the generator ID is used to identify different modules that use the same PMID. For example, the TPBPUTL PMID identifies BASE24-billpay utilities. Generator IDs of Cnfg, Grd, and Xport, further identify whether the event message was generated specifically by the configuration utilities, Guardian utilities, or transport utilities, respectively.

PCI Data Security Standards



The BASE24 security best practices are based on industry security guidelines, which include Payment Card Industry (PCI) Data Security Standards (DSS) version 1.1. The PCI storage and display requirements that affect the handling of BASE24 event messages are as follows:

- Sensitive authentication data cannot be stored subsequent to authorization, even if encrypted. Sensitive authentication data is defined as the full magnetic stripe, the CVC2/CVV2/CID, and the PIN/PIN Block.
- The personal account number (PAN) must be masked when displayed. At most, the first six and last four digits may be displayed.
- The cardholder name, service code, and expiration date must be protected if stored in conjunction with the PAN.

The DIRECTIVE field on the Log Message Configuration File (LMCF) screen allows users to modify the way in which the event logging procedures handle the sensitive authentication data and cardholder information in an event message. Refer to [“Using the MASK and MAX Directives”](#) in this section for information on the values that can be specified in the DIRECTIVE field and examples of how the different values affect the way in which message text is logged.

Log Message Configuration File (LMCF)

The LMCF contains records for all user-modified event messages and is read at initialization by all BASE24 application processes. The routing code, severity, text, directive, and PAN digits value of an event message can be altered by a BASE24 operator by modifying the Log Message Configuration File (LMCF). The LMCF also allows customers to add site-specific information to BASE24 event messages. The Online Maintenance File (OMF) provides a detailed audit trail of every LMCF record that is added, updated, or deleted.

BASE24 processes use PMIDs to retrieve modified event message records from the LMCF for each type of process that can read the file. The LMCF contains changes made to event messages, in the form of records, that are read by all of the BASE24 processes and XPNET processes that route event messages.

Using the Log Message Configuration File

The LMCF screen enables you to modify the routing code, severity, text, directive, and PAN digits value of any standard format BASE24 event message. The function of the screen is one of file maintenance: to add, read, update, and delete event messages in the LMCF. This subsection describes the LMCF screen and the procedure for modifying an LMCF record.

LMCF Screen

The LMCF screen is shown below, followed by descriptions of its fields.

```
BASE24-LMCF  LOG MESSAGE CONFIG      LLLL      YY/MM/DD  HH:MM  01 OF 01

PRODUCT MODULE ID:
MESSAGE NUMBER:

ROUTING CODE:
SEVERITY CODE:
MESSAGE FORMAT:


DIRECTIVE:

PAN DIGITS:


*****
                DESTINATION:                LOGICAL NETWORK ID:
                F12-HELP
```

PRODUCT MODULE ID — Specifies the product module ID (PMID) for the process for which message information is to be read, updated, or deleted. A list of valid product module IDs is included in section 2.

Field Length: 1–16 alphanumeric characters
Required Field: No
Default Value: No default value

MESSAGE NUMBER — Specifies the message number for which information is to be read, added, updated, or deleted. This number is the number used by the application process for identifying a specific message.

Field Length: 1–4 numeric characters
Required Field: No
Default Value: No default value

ROUTING CODE — A routing code to be used when the message is generated by the application process indicated in the **PRODUCT MODULE ID** field. This field allows you to override the normal routing code used by the application process for its messages. The default routing codes for all BASE24 processes, by product, are as follows:

- 0 = Shared processes
- 1 = BASE24-atm processes
- 2 = BASE24-pos processes
- 3 = BASE24-teller processes
- 8 = BASE24-from host maintenance processes

Field Length: 1–4 numeric characters
Required Field: No
Default Value: No default value

SEVERITY CODE — The severity code to be used when the message is generated by the application process indicated in the **PRODUCT MODULE ID** field. This field allows you to override the normal severity code associated with a particular message. Valid values are as follows:

- 0 = Information
- 1 = Warning
- 2 = Error
- 3 = Critical

This field can also be set to the value 9, which causes this event message to be suppressed by the SKEL logging utilities. When this value is used, the message is not logged to any logging location.

Field Length: 1 alphanumeric character
Required Field: No
Default Value: No default value

MESSAGE FORMAT — The message text to be used when this event message is generated. This field allows you to override the normal message text associated with a particular message. The maximum length for the text portion of an event message is 342 characters.

Warning: Exercise extreme care when replacing the message text. Unpredictable results occur when the sequence and number of special formatting characters are altered.

Field Length: 40 alphanumeric characters
Occurs: 3 times
Required Field: No
Default Value: No default value

DIRECTIVE — Indicates whether the specified event message is to be logged to a specified maximum length, logged with all or selected parameters masked with asterisks, or logged without modification. Valid values are as follows:

MASK = Log the event message with all or the specified parameters masked by asterisks.
MAX = Log the event message for a specified maximum length and truncate the remainder of the message.

If this field is left blank, the entire event message text is logged and none of the parameters are masked.

Refer to [“Using the MASK and MAX Directives”](#) in this section for additional information on this field.

Field Length: 28 alphanumeric characters
Required Field: No
Default Value: Blanks

PAN DIGITS — A 3-digit value that indicates how to mask the PAN and similar parameters in the event message that corresponds to this LMCF record. If this field contains valid numeric values, those values override the masking specified by the AFT-PAN-DIGITS parameter in the Logical Network Configuration File (LCONF). If this field contains all blanks or the value ULF (use LCONF file), the AFT-PAN-DIGITS parameter in the LCONF is used to mask the parameter in the event message. The table shown below defines each of the digits in this field and lists the valid numeric values for each digit.

Digit Number	Description	Valid Numeric Values
1	The maximum number of left most digits to be displayed unmasked.	0–6
2	The minimum number of digits to be masked.	0–9
3	The number of rightmost digits to be displayed unmasked.	0–4

Note: This field must be set to three valid numeric values, all blanks (default), or the value ULF (use LCONF file).

Field Length: 3 alphanumeric characters
Required Field: No
Default Value: Blanks

Refer to [“Using the MASK and MAX Directives”](#) in this section for an example of how this field is used.

Function Keys

The standard BASE24 function keys are used on the LMCF screen. These function keys are described in the *BASE24 Base Files Maintenance Manual*. The Help screen also lists the function keys applicable to this screen. To display the Help screen, press the **F12** key.

Modifying an LMCF Record

To retrieve an LMCF record to add, modify, or delete characteristics of a particular event message, perform the following steps:

1. Access the LMCF screen, using one of the following methods:
 - Select LMCF from the Operator Product Menu.
 - Enter LMCF in the FILE DESTINATION field at the bottom of a CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key.

The LMCF screen is displayed.

2. Enter the appropriate values in the PRODUCT MODULE ID and the MESSAGE NUMBER fields on the LMCF screen. These are the only fields required to read a record.
3. Press the **F2** key to read the record.
4. To modify an LMCF record, add or change the appropriate information and then use one of the following functions:
 - To add a record, press the **F3** key.
 - To modify an existing record, press the **F5** key.
 - To delete a record, press the **F4** key.

When a BASE24 application process generates an event, the SKEL library procedures reference the LMCF records in memory for the process to determine whether the event message requires modification. If an LMCF record is located in memory, the procedures modify the event message as indicated before sending it to the EMS collector process. Otherwise, the event message is sent to the collector process without modification. Because LMCF records are read into memory by the processes using them, updates to the LMCF do not take effect until the processes are warmbooted or restarted.

Using the MASK and MAX Directives

The DIRECTIVE field on the Log Message Configuration File (LMCF) screen is used to provide instructions to the SKEL event logging procedures on how to mask or eliminate the sensitive authentication data and cardholder information in an event message before it is sent to the EMS collector process.

The following section describes the purpose and syntax of the MASK and MAX directives. Also provided are examples showing different directive values on the LMCF screen and the event message text that is logged as a result of those values.

MASK Directive

The MASK directive instructs the SKEL event logging procedures to mask all or selected parameters in an event message with asterisks. A note, ****PCI MASKED****, is appended to the event message text to inform the operator that the event message was masked because of this directive.

The syntax of the MASK directive is shown below:

```
MASK [parameter number], [parameter number], ...[parameter number]
```

The *parameter number* is an optional value that specifies the ordinal number of each parameter in the event message format template that is to be masked. Multiple parameter numbers are comma separated. For example, MASK 1, 3 indicates that the first and third parameters in the event message format template are to be masked.

If a parameter number is specified, the event logging procedures mask the specified parameter if it uses one of the formatting characters shown below. If no parameter numbers are specified with the MASK directive, the event logging procedures mask every parameter in the event message that uses one of the formatting characters shown below.

- * Variable-length string of any length
- \? Variable-length string of any length
- \\ Fixed-length string of any length

If a parameter uses one of the formatting characters shown above and is 13–19 characters in length, it is masked according to the PAN DIGITS field on the LMCF screen, unless that field contains the value ULF (use LCONF file) or blanks. If the PAN DIGITS field is set to ULF or contains blanks, the parameter in the event message is masked according to the AFT-PAN-DIGITS parameter in the Logical Network Configuration File (LCONF). The specified digits are replaced with asterisks. It is assumed that a parameter of this length is a card number or account number.

If the length of a parameter that is to be masked is less than 13 characters or greater than 19 characters, every character is replaced with asterisks.

Example 1:

In this example, a BASE24 program defines an event message using the following format template, then calls the SKEL event logging procedure LOGMESSAGE^.

```
wlform( m3, \s - Invalid data from card verify utilities, "  
      ', ' "bank \\\\", PAN \?, exp date \\\\" )  
  
call logmessage^( 3288, rte^cde^g, @m3, net.myname,  
                  log^error^l, @auth^cde^g, @fiid^g,  
                  @trk2.pan, pan^lgth, @trk2.exp^dte );
```

Event message 3288 has the following record in the LMCF.

```
BASE24-LMCF  LOG MESSAGE CONFIG      LLLL      YY/MM/DD  HH:MM  01 OF 01

PRODUCT MODULE ID:  PSSPDH
MESSAGE NUMBER:     3288

ROUTING CODE:
SEVERITY CODE:
MESSAGE FORMAT:

DIRECTIVE:           MASK 3, 4

PAN DIGITS:

*****
DESTINATION:         LOGICAL NETWORK ID:
F12-HELP
```

The DIRECTIVE field contains the value MASK followed by the number 3, a comma, and the number 4. The numbers 3 and 4 identify the third and fourth parameters in the event message.

The event logging procedures mask the third parameter with asterisks as defined by the AFT-PAN-DIGITS parameter in the LCONF because the parameter contains one of the formatting characters listed previously and is 13–19 characters in length. (If the PAN DIGITS field on this screen contained valid numeric values instead of blanks, it would override the value of the AFT-PAN-DIGITS parameter in the LCONF.)

The event logging procedures mask every character in the fourth parameter with asterisks because it contains one of the formatting characters listed previously and it is not 13–19 characters in length.

The event logging procedures also append a note indicating the event message was masked because of the MASK directive. Then the event logging procedures send the following event message text to the EMS collector process:

```
S1A^TERM1 - Invalid data from card verify utilities, bank BNK1,
PAN 654321*****1234, exp date **** **PCI MASKED**
```

Example 2:

This example uses the same event message that was used in the first example, but it uses a different directive value.

```

BASE24-LMCF  LOG MESSAGE CONFIG          LLLL          YY/MM/DD  HH:MM  01 OF 01

PRODUCT MODULE ID:  PSSPDH
MESSAGE NUMBER:     3288

ROUTING CODE:
SEVERITY CODE:
MESSAGE FORMAT:

DIRECTIVE:          MASK

PAN DIGITS:         484

*****
DESTINATION:        LOGICAL NETWORK ID:
F12-HELP

```

The **DIRECTIVE** field contains only the value **MASK**, indicating that all of the parameters should be masked, provided that each contains one of the formatting characters listed previously.

The event logging procedures mask every character in the second and fourth parameters with asterisks because they contain the formatting characters listed previously and they are not 13–19 characters in length.

The event logging procedures mask the third parameter with asterisks as defined by the **PAN-DIGITS** field on the LMCF screen because the parameter contains one of the formatting characters listed previously and is 13–19 characters in length. (If the **PAN DIGITS** field on this screen had been set to **ULF** or contained all blanks, the parameter in the event message would be masked as defined by the **AFT-PAN-DIGITS** parameter in the **LCONF**.)

The event logging procedures also append a note indicating the event message was masked because of the **MASK** directive. Then the event logging procedures send the following event message text to the EMS collector process:

S1A^TERM1 - Invalid data from card verify utilities, bank ****,
PAN 6543*****1234, exp date **** **PCI MASKED**

Note: The first parameter in the event message is not masked because it does not contain one of the formatting characters listed previously.

MAX Directive

The MAX directive instructs the SKEL event logging procedures to log an event message for a specified maximum length. The remainder of the event message text is truncated. A note, ****PCI TRUNCATED****, is appended to the event message text to inform the operator that the event message was truncated because of the MAX directive.

The MAX directive is typically used when an application logs an entire inter-process message or data record that may contain sensitive authentication data or cardholder information. Only the first few bytes of the inter-process message or data record should be logged, which is enough to provide diagnostic value, but not enough to contain sensitive information.

The syntax of the MAX directive is shown below:

MAX length

The *length* is a required value that specifies the maximum length of the event message text. The event logging procedures log only the first *length* bytes of the event message text. Valid values are 0–342. If 0 is entered, only the note ****PCI TRUNCATED**** is logged.

Example:

In this example, a BASE24 program logs the entire contents of a BASE24-pos Standard Internal Message (PSTM) using the SKEL event logging procedure LOG^MESSAGE.

```
call log^message( 2205, rte^cde^g, pstm, max^logger^1,  
                 net.myname, 2 )
```

Event message 2205 has the following record in the LMCF.

```
BASE24-LMCF  LOG MESSAGE CONFIG      LLLL      YY/MM/DD  HH:MM  01 OF 01
```

```
PRODUCT MODULE ID:  PSSPDH
MESSAGE NUMBER:      2205
```

```
ROUTING CODE:
SEVERITY CODE:
MESSAGE FORMAT:
```

```
DIRECTIVE:           MAX 20
```

```
PAN DIGITS:
```

```
*****
```

```
DESTINATION:          LOGICAL NETWORK ID:
F12-HELP
```

The DIRECTIVE field contains the value MAX followed by the number 20. Therefore, the event logging procedures send only the first 20 bytes of event message 2205 to the EMS collector process, plus a note indicating the event was truncated, as shown below. Any sensitive authentication data or cardholder information that would have otherwise been logged is suppressed.

```
0200020000S1A^TERM1  **PCI  TRUNCATED**
```

Event Message Format

Event messages are displayed with a subsystem identifier, event message number, and timestamp. Event messages with a severity of 3 (critical), can be displayed with emphasis (e.g., in reverse video) if your Event Management Service (EMS) subsystem is configured to do so.

For procedural BASE24 products, there is an exception to the standard event message format. Event messages generated by the SKEL utilities, which are bound into BASE24 procedural product modules, are displayed as if they were produced by the product modules with which they were bound. The only difference between the messages generated by the SKEL utilities and the messages generated by the product module is that the product module ID displayed is SKEL, SKEL-I, SKEL-U, or SKEL-T. When a SKEL utilities product module ID is displayed, the event message documentation is found in the SKEL utilities file.

The above exception does not apply to the BASE24-telebanking product, which is object-oriented.

Event Message Content

The actual formats of the standard event messages are different depending on the method by which the event message is displayed. These formats are described below. However, all standard event messages contain the following information:

- Date, in YY/MM/DD format, the event message was generated.
- Time of day, in hh:mm:ss.ttt format, the event message was generated.
- Manager. The fully qualified process pair directory (PPD) name of the process that started the process that generated the message.
- Subsystem ID. For BASE24 products, the product module ID (PMID) is displayed as part of the subsystem ID. For example, an event message identifying the PMID for the BASE24-atm Authorization process contains the subsystem ID of ACL.ATAUTH.1000, where ATAUTH is the PMID.
- Event message number.
- Generator ID. This identifier provides the symbolic name of the message generating process or a hard-coded 8-character string defined and passed by the process to further identify the generator of the message. The generator ID is typically prefaced with the text "FROM:" in the event message.

- Subject. The subjects with which the event is concerned. The subject typically contains a symbolic process name and is prefaced with the text “RE:”.
- Message text.

BASE24 Example

The following is an example of how event message 0024 is displayed for the BASE24-teller Settlement Initiator process. In this example, the PMID is TRSETL.

YY/MM/DD HH:MM:SS.TTT \PRO1.\$P1SE	ACI.TRSETL.1000	24
FROM: P1A^TSETL	RE: P1A^TSETL	
FIID TBK1 has an illegal cutover time of 2330. End cutover must be 1/2 hour before network cutover. Defaulting to network cutover		

NET24-XPNET Example

The following is an example of how event message 6235 is displayed for the XPNET process when logged to a CRT or a printer. In this example, the PMID for the XPNET process, XPNET, is expressed as the integer 328. The version number indicates that the message was generated by XPNET release 2.1.

YY/MM/DD HH:MM:SS.TTT \PRO1.\$P1AN	ACI.328.2100	06235
Abnormal lines L1A^HPDH		

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2: How Event Messages Are Documented

BASE24 event message documentation explains the text of each event message, identifies the cause of each event message, and provides a remedy for the condition identified by each event message. This documentation is provided in HP NonStop edit files and is intended to be printed as needed. You can then keep your printed event message documentation in a handy location for easy access.

This section explains how BASE24 event messages are documented. It includes the following topics:

- Getting started
- Event message file organization and format
- Documentation organization and format
- Special use documentation files
- Where event messages are documented

Getting Started

This subsection explains what you need to get started using the event message documentation. It provides an explanation of how to find the event message documentation, recommendations for using the event message documentation, and an explanation of how pages within the event message documentation files are arranged.

Finding Event Message Documentation

BASE24 event message documentation is located in files on the HP NonStop subvolumes called BAxxLOGM and OKxxLOGM, where xx is the number of the current release, or on switch-specific subvolumes. Files on the BAxxLOGM subvolume contain event message documentation for procedural BASE24 products. Files on the OKxxLOGM subvolume contain event message documentation for object-oriented BASE24 products. Files on both subvolumes are organized alphabetically by product module IDs (PMIDs). Each file contains all of the event messages associated with a particular PMID. Switch documentation is found in a file called xxxxLOGM on the switch-specific subvolume, where xxxx is the name of the switch. Refer to the topic “Interface PMIDs” later in this section for more information about switch-specific subvolumes and event-message file names.

PMIDs are the key for getting from an event message to its documentation. You should become familiar with the PMIDs in order to access the appropriate event message documentation. To access event message documentation, you must first identify the PMID that generated the event message. The PMID then leads to the file containing the documentation for the process. For a list of specific files and the event message documentation they contain, refer to the topic “Where Event Messages are Documented.”

The printed TGAL output for the same files included on the BAxxLOGM, OKxxLOGM, and switch-specific subvolumes are also included on the ACI Product Documentation CD-ROM. Instructions for accessing event message documentation from the CD-ROM are provided in section 3.

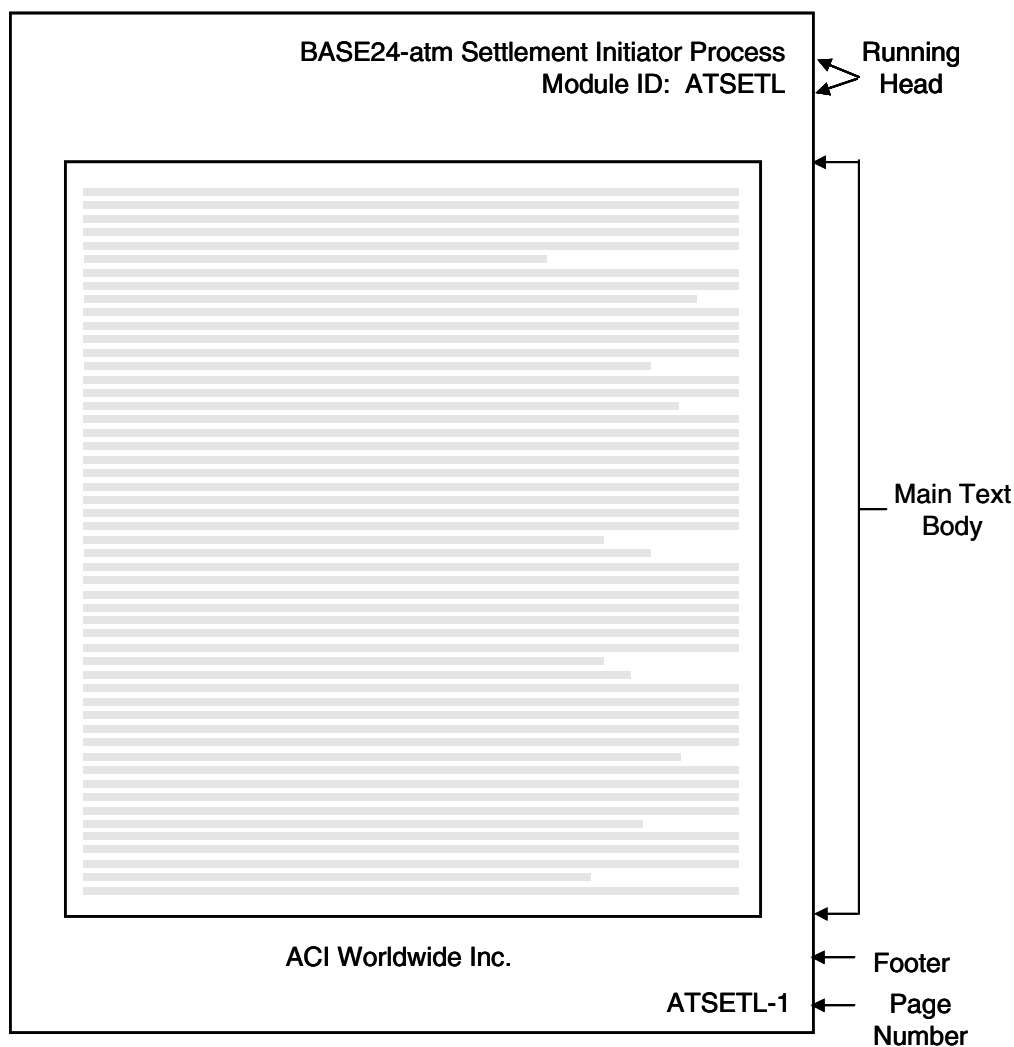
Printing Event Message Documentation

Once you become familiar with the way the event message documentation is set up, ACI recommends that you print all of the event message documentation you require and place them in alphabetical order, by PMID, in a binder. The printed

event messages can then be kept in a central location so operators can use them as needed. For instructions on printing event message documentation files, refer to section 3.

Using the File Page Layout

The pages within the printed event message documentation are arranged to assist you in organizing your event message documentation. The following diagram illustrates an example of the printed event message page layout:



Pages in printed event message documentation consist of the following components:

Running Head — Identifies the BASE24 product, process, and corresponding PMID. The running header serves as a marker to indicate which event message documentation file you are currently viewing.

Main Text Body — Contains the text of the page.

Footer — Identifies the date, in MM/DD/YY format, the file was last updated. This allows you to monitor changes to the documentation.

Page Number — Consists of the PMID followed by a number. The page number is especially helpful when you are trying to locate specific event messages from your printed documentation. Once event message documentation files are printed and placed together in binders, you can simply flip through the pages, using the numbering scheme at the bottom of the page to locate the appropriate PMID. This page numbering scheme also makes it easy to distinguish where documentation for one process ends and documentation for the next process begins.

Event Message File Organization and Format

The event message files on the HP NonStop subvolumes use a format that includes the following components:

- Introduction
- Document history
- Event message documentation

Note: Each file on the BAxxLOGM and OKxxLOGM subvolumes, where *xx* is the number of the current release, also includes TGAL formatting commands. TGAL is a HP NonStop utility used to format text files for printing. The TGAL commands are required for printing files using TGAL. These commands can easily be recognized because they begin with a backslash (\). When viewing an event message documentation file, you can ignore these commands. For more information on TGAL, refer to the HP NonStop *TGAL Manual*.

Introduction

Each event message file begins with an introduction. The main purpose of the introduction is to identify the process, program, or module for which event messages are documented in the file. The introduction also includes copyright information.

In addition, the introduction serves to notify you of any deviations in the file from normal event message documentation. For example, in some cases, a single process may be associated with multiple PMIDs. In this situation, the event message documentation subvolume contains a separate file for each different PMID. Therefore, a process such as the ISO Host Interface process may have several files residing on the event message subvolume. The ISO Host Interface process may have a file for BAHISO for its Base event messages, ATHISO for its BASE24-atm event messages, FHMHISO for its BASE24-from host maintenance event messages, PSHISO for its BASE24-pos event messages, TBHISO for its BASE24-telebanking event messages, and TLRHISO for its BASE24-teller event messages. You are made aware of these special conditions in the introduction.

Document History

As event messages are added, deleted, and modified in the BASE24 system, you need to be aware of what has changed. Each event message file contains a document history section to serve this purpose. Each time a change is made to the file, the document history section is revised. The document history section identifies each set of changes made to the file, the dates they were made, and the initials of the person who made the change.

The document history section notifies you of updates to the specific file and tracks changes occurring in each file. You can determine if you have the most recent version of the file based on the information contained in the document history section. A description column identifies the changes and includes a brief explanation of each change.

Event Message Documentation

The next section of the event message file contains the actual documentation for the event messages generated by the specified process, program, or module. For more information on the event message documentation structure, refer to the “Documentation Organization and Format” topic.

Documentation Organization and Format

Standard event messages are organized within files in numerical order by event message number. These messages include an event message number and severity code. Nonstandard event messages are arranged in alphabetical order based on the first letter of the message text. These messages do not include an event message number or severity code. Individual event messages in the files are separated by a string of asterisks (*).

The example below is from an event message documentation file for the BASE24-atm Authorization process (ATAUTH). The fields in the file are described following the example.

NUM: 0128
SEV: 2
TXT: KEYA entry not found for *****, GROUP \\\, EXP
TXT: DATE \\\

VAR: *'s = Either "Bank \\\\" where \\\ = FIID or "Prefix *'s"
VA2: where *'s = the prefix.
VAR: \\\ = Group ID being used in combination with record type as
VA2: the key into the KEYA file.
VAR: \\\ = Expiration date of the card being processed, to be
VA2: matched against the begin and end date ranges in the KEYA.

CAUSE:
A transaction was received with a prefix/group/expiration date or
FIID/group/expiration date combination that did not match with any Key
Authorization File (KEYA) entries.

REMEDY:
Add or modify the appropriate KEYA record to support this combination
and send a WARMBOOT ALL (9501****) command to the Authorization
process.

NUM: 0129
SEV: 3
TXT: POFST/PVV value not located on the CAF when performing a
TXT: PIN change transaction

CAUSE:
A PIN change transaction was requested, but the offset for the card
involved in the transaction is not maintained in the Cardholder

Authorization File (CAF).

REMEDY:

Correct PIN verification data and send a warmboot all command to the Authorization process.

If PIN verification data is being maintained in the Institution Definition File (IDF), change the PIN Offset Location indicator on screen 2 of the IDF to a 2 (Offset on CAF) and send the Authorization process a warmboot for this FIID. If PIN verification data is being maintained in the Card Prefix File (CPF), change the PIN Offset Location indicator on screen 2 of the CPF to a 2 (Offset on CAF) and send the Authorization process a warmboot all command.

If PIN verification data is being maintained in the IDF, change the PIN Offset Location indicator on screen 2 of the IDF to a 2 (Offset on CAF). If PIN verification data is being maintained in the CPF, change the PIN Offset Location indicator on screen 2 of the CPF to a 2 (Offset on CAF). Send the Authorization process a warmboot all command.

NUM: 0130

SEV: 2

TXT: PVN for clear PINs cannot be calculated using Identikey

CAUSE:

A PIN change transaction was requested for a clear PIN using the Identikey algorithm. No support exists for the Identikey algorithm in software. The Identikey algorithm is a proprietary algorithm.

REMEDY:

If the PIN change transaction is desired using clear PINs, the PIN verification method must be changed.

NUM — The number identifying the event message. This field is used only with event messages generated in standard event message format.

SEV — A code indicating the severity of the event message. This field is used only with event messages generated in standard event message format. The severity code denotes the urgency with which to act on the information identified in the event message. Valid values are as follows:

0 = Informational — General system information; not used to denote problems, but rather to convey helpful information on the operation of the system. Normally, no action is required for these messages. A typical message in this category might be that a process has just completed initialization successfully.

1 = Warning — Indicates potential problems on which you may have to act. No action is warranted at this time, but you should watch the log for further developments. A typical message in this category might be that the system has suspended a station for a given interval.

Note: Text messages sent to the log from the Network Control Supervisor (NCS) or NCPCOM screens are included in this category.

2 = Error — Indicates some type of processing or hardware error that is not critical to normal processing; action is usually required to correct the error condition. A typical message in this category might be that a station or line has been marked down by the system.

3 = Critical — Indicates that a processing or hardware error has occurred that requires immediate action in order for processing to continue. A typical message in this category might be that the system is unable to read a record in a required file during initialization.

TXT — Identifies the text associated with the event message. If the text is longer than one line, TXT is placed at the beginning of each new line.

VAR — A description of the variable items (if any) in the event message text.

VA2 — Identifies subsequent lines of the description for a variable. If the description of the variable associated with the event message is longer than one line, the subsequent lines are identified by the literal VA2.

CAUSE — A description indicating the reason the event message was generated.

REMEDY — A description of the action to be taken to correct the situation that caused the message to be generated.

Special Use Documentation Files

Along with event message documentation files, two types of special use documentation files reside on the BAxxLOGM or OKxxLOGM subvolumes: the README and print files. The README file resides on the BAxxLOGM subvolume and is included to provide you with basic information about the event message documentation. Print files reside on either subvolume and are designed to assist you in printing the event message documentation you require. These special use files are described below.

README File

The README file provides basic information about the event message documentation such as instructions on how the subvolume is organized and how to print event message documentation, as well as general information to help you understand the format and organization of the event message documentation. You can print the README file in the same manner as any other file on the subvolume. For information on printing files, refer to section 3.

Print Files

Another set of files on the HP NonStop subvolume are print files. Print files are organized by product and allow you to print all of the event message documentation associated with a particular product. Print files are named XPRINTxx, where xx identifies the product for which event message documentation is produced using the print file. The xx variable can have the following values:

AT	=	BASE24-atm
BA	=	Base
BP	=	BASE24-billpay
CO	=	Core
FH	=	BASE24-from host maintenance
IE	=	Integrated endpoint
KN	=	Kernel
MA	=	BASE24-mail
PS	=	BASE24-pos
TB	=	BASE24-telebanking
TP	=	BASE24-billpay
TR	=	BASE24-teller

You can also modify print files to include the files you want to print out. For information on modifying print files, refer to section 3.

Where Event Messages Are Documented

This subsection identifies the files where the event messages currently documented for BASE24 product modules are located. The following tables are divided by product. Each table contains a list of the product modules associated with the specified product and the corresponding files, identified by product module ID (PMID), where the event message documentation resides. For BASE24 product modules other than interfaces, the PMID exactly matches the corresponding event message documentation file name. In the case of interfaces, all PMIDs begin with “SW,” and most are followed by the first three or four characters of the corresponding event message documentation file name. Each of the interface event message documentation file names ends in “LOGM,” which is not part of the PMID.

Each entry in this subsection is formatted in the following manner:

File Name or Subvolume/File Name — Identifies the file on the subvolume where event messages for a particular product module reside, or identifies the subvolume and file where event messages for a particular interface module reside.

Product Module Name — Identifies the product module (e.g., Settlement Initiator, Authorization, Router, etc.) within BASE24 that generates the event messages described in the corresponding file.

The *NET24-XPNET Event Management Manual* identifies the files where the event messages currently documented for the XPNET subsystem are located.

BASE24 Base PMIDs

The following PMIDs identify files that contain the documentation for the event messages generated by BASE24 Base product modules. All of these files reside on the BAxxLOGM subvolume.

File Name	Product Module Name
BAHISO	ISO Host Interface process (Base module)
BAMCUTLS	Multiple Currency utilities
BAREFR	Refresh process

File Name	Product Module Name
BASUPERX	Super Extract process
SECUTILS	BASE24 Security utilities

BASE24-atm PMIDs

The following PMIDs identify files that contain documentation for the event messages generated by BASE24-atm product modules. All of these files reside on the BAxxLOGM subvolume.

File Name	Product Module Name
AT1000	Diebold 1000-series Device Handler process
AT24V7	IBM 3624 (Version 7) Device Handler process
AT3624	IBM 3624 (Version 8) Device Handler process
AT4730	IBM 4730-series Device Handler process
AT5070	NCR 5XXX Device Handler process
AT910	Diebold 910/911 Device Handler process
AT912	Diebold 912 Device Handler process
ATAUTH	Authorization process
ATFUJ	Fujitsu Device Handler process
ATFUJ7	Fujitsu 7000-series Device Handler process
ATHISO	ISO Host Interface process (BASE24-atm module)
ATMCUTLS	Multiple Currency utilities
ATSETL	Settlement Initiator process
ATTIDL	Tidel Device Handler process
ATTRIT	Triton Device Handler process
ATUTILS	Utilities

BASE24-billpay PMIDs

The following PMIDs identify files that contain documentation for the event messages generated by BASE24-billpay product modules. All of these files reside on the OKxxLOGM subvolume.

File Name	Product Module Name
TPBPACHC	ACHclear program
TPBPARCH	Archive program
TPBPBBLG	Bank Billing and Bank Day programs
TPBPBKAV	Bank Advice process
TPBPBLG	Billing utilities
TPBPBPSV	Billpay Server process
TPBPCBLG	Customer Billing program
TPBPCLUP	Cleanup program
TPBPDBUT	Database utilities
TPBPRMTG	Remittance Generator process
TPBPSTI	Scheduled Transaction Initiator process
TPBPTXUT	Transaction utilities
TPBPUTL	Utilities

BASE24-from host maintenance PMIDs

The following PMIDs identify files that contain documentation for the event messages generated by the BASE24-from host maintenance product module. All of these files reside on the BAxxLOGM subvolume.

File Name	Product Module Name
FHM	From Host Maintenance process
FHMHISO	ISO Host Interface process (BASE24-from host maintenance module)

BASE24-pos PMIDs

The following PMIDs identify files that contain documentation for the event messages generated by BASE24-pos product modules. All of these files reside on the BAxxLOGM subvolume.

File Name	Process Name
PSAUTH	BASE24-pos Device Handler/Router/Authorization process (BASE24-pos Authorization module)
PSCRTA	CRT Device Handler process CRT Administration Device Handler process CRT Authorization Device Handler process
PSHISO	ISO Host Interface process (BASE24-pos module)
PSHYPRCM	Hypercom Device Handler process
PSMCUTLS	Multiple Currency utilities
PSMHI	Merchant Host Interface process
PSNCRNDP	NCR NDP Device Handler process
PSRAEMV	Router/Authorization EMV support module
PSROUTER	BASE24-pos Device Handler/Router/Authorization process (BASE24-pos Router module)
PSSETL	Settlement Initiator process

File Name	Process Name
PSSPDH	BASE24 Standard POS Device Handler process
PSV1DH	VISA I Device Handler process
PSV2DH	VISA II Device Handler process

Note: Event messages for the APACS 30/40 POS Device Handler process are documented in the ***BASE24-pos APACS 3040 Device Handler User's Guide***.

BASE24-teller PMIDs

The following PMIDs identify files that contain documentation for the event messages generated by the BASE24-teller product modules. All of these files reside on the BAxxLOGM subvolume.

File Name	Product Module Name
TLRHISO	ISO Host Interface process (BASE24-teller module)
TRAUTH	Authorization process
TRSETL	Settlement Initiator process

Core PMIDs

The following PMIDs identify files that contain documentation for the event messages generated by object-oriented core product modules. This includes the BASE24 Customer Service and BASE24-telebanking products. All of these files reside on the OKxxLOGM subvolume.

File Name	Product Module Name
ISCOACAC	Account Access object
ISCOACCT	Account object
ISCOACPC	Authorization Combined Positive Customer object
ISCOACU	Authorization Combined Utility object

File Name	Product Module Name
ISCOASAC	Authorization Security Access object
ISCOCSAC	Customer Access object
ISCOCSI	Customer Service Interface object
ISOCUST	Customer object
ISCOEOPP	End-of-Period Process object
ISCOFI	Financial Institution object
ISCOFIAC	Financial Institution Access object
ISCOHIST	Transaction History object
ISCOIMGR	Interface Manager object
ISCOIMS	Interface Master Services object
ISCOIMTE	Internal Message Translation Endpoint Class Handler object
ISCOIMTT	Internal Message Translation Interface object
ISCOMC	Multiple Currency object
ISCOREFR	Refresh process
ISCORTE	Router object
ISCOTSEC	Transaction Security object
ISCOTXN	Transaction object
ISCOVRUE	Voice Response Unit Endpoint Class Handler object
ISCOVRUM	Voice Response Unit Master object
ISCOVRUT	Voice Response Unit Interface object
ISCOYANG	Yang object

Integrated Endpoint PMIDs

The following PMIDs identify files that contain documentation for the event messages generated by object-oriented integrated endpoint product modules. All of these files reside on the OKxxLOGM subvolume.

File Name	Product Module Name
ISIECTXM	Context Manager object
ISIEDEST	Destination object
ISIEEPCS	Endpoint Class Handler Services object
ISIEGENE	Generic Endpoint Class Handler object
ISIEMISO	Message ISO object

Interface PMIDs

The following table identifies the files that contain documentation for event messages generated by interface product modules. Each of these files resides on an interface-specific subvolume.

Subvolume/File Name	Product Module Name
SW _{xx} AMXI.AMXILOGM	American Express International Interface process
SW _{xx} AXCI.AXCILOGM	American Express Interface process
SW _{xx} BABI.BABILOGM	BIC ISO Interface process
SW _{xx} BIC.BICLOGM	BIC Interface process
SW _{xx} BNET.BNETLOGM	Banknet Interface process
SW _{xx} MDS.MDSLOGM	MDS ISO Interface process
SW _{xx} PISO.PISOLOGM	Plus ISO Interface process
SW _{xx} VISA.VISALOGM	VisaNet ISO Interface process

The following table identifies the PMIDs associated with each of the event message documentation files named above.

File Name	PMID
AMXILOGM	SWAMXI
AXCILOGM	SWAXCI
BABILOGM	SWBICI
BICLOGM	SWBIC
BNETLOGM	SWBNET
MDSLOGM	SWMDS
PISOLOGM	SWPISO
VISALOGM	SWVISA

Note: Event messages for the EPSNet and LINK Interchange Interface processes are documented in the **BASE24 EPS-Net Interchange Interface Manual** and the **BASE24 LINK ISO Interchange Interface Manual**, respectively.

Kernel PMIDs

The following PMIDs identify files that contain documentation for the event messages generated by object-oriented kernel product modules. All of these files reside on the OKxxLOGM subvolume.

File Name	Product Module Name
ISKNANS	Application Network Services object
ISKNCLAS	Class object
ISKNEMTB	Extended Memory Table Build utility
ISKNEVT	Event object
ISKNLOGP	Log Permission object
ISKNPCTL	Process Control object
ISKNTRC	Trace object
ISKNXPXN	XPNET Interface object

3: Using Event Message Documentation

This section explains how to access event message documentation and special use files. It also describes several options you have for printing event message documentation.

Printing and Assembling Event Message Documentation

ACI recommends that you print and assemble all of the event message documentation you require. The following instructions describe one way to print and assemble your event message documentation.

1. Identify the product module IDs (PMIDs) associated with the product modules that generated the event messages you want to print. The PMIDs of each message can be determined from the messages as they are displayed on the log.
2. Log on to the HP NonStop using your assigned logon and password. After successful logon, a TACL prompt is displayed.
3. Enter the VOLUME command from the TACL prompt to move to the HP NonStop subvolume where the event message documentation is located. If you do not know the subvolume where the documentation files are located, refer to section 2 for a complete list of PMIDs and the subvolumes on which they reside. In this example, assume the documentation is located on the BA60LOGM subvolume.

```
VOLUME $REL60.BA60LOGM
```

4. List the files on this subvolume by entering the following command from the TACL prompt.

```
FILES
```

BASE24 event messages are documented on this subvolume according to the product module that generated each message and are organized alphabetically by PMID. For example, all of the event messages generated by the BASE24-pos Settlement product module are documented in the PSSETL file.

The following is an example of the output displayed when this command is entered.

```
$REL60.BA60LOGM
```

AT1000	AT24V7	AT3624	AT4730	AT5070	AT910	AT912	ATATMCUT
ATAUTH	ATFUJ	ATFUJ7	ATHISF	ATHISO	ATSETL	ATTIDL	ATTRIT
ATUTILS	BABAMCUT	BABAUTIL	BAHISF	BAHISO	BAREFR	BASUPERX	BATAR
FHM	FMHISF	FMHISO	PSAUTH	PSCRTA	PSHISF	PSHISO	PSHYPRCM
PSMHI	PSMTTL	PSNCRNDP	PSPSMCUT	PSROUTER	PSSETL	PSSPDH	PSV1DH
PSV2DH	README	SECUTILS	TBHISO	TLRHISO	TRAUTH	TRSETL	XPRINTAT
XPRINTBA	XPRINTPS	XPRINTTR					

5. Locate the files on the subvolume by the product module ID (PMID) corresponding to the product modules for which you want to print event message documentation. If you do not know the PMIDs, refer to section 2 for a complete list of PMIDs and their corresponding product module names.

If you want to print all of the event messages for a particular product, use the print files provided on the subvolume. For more information on print files refer to the “Printing Multiple Files” topic in later in this section.

You can modify the print files residing on the subvolume to meet your specific requirements. The following example shows how to modify a print file:

- a. Use a HP NonStop editor to edit the appropriate print file on the subvolume. In this example, the XPRINTAT file, which prints all of the event message files for BASE24-atm product modules, requires modification. When you edit the XPRINTAT file, the following is displayed:

```
\in at1000
\new odd
\in at2300
\new odd
\in at24v7
\new odd
\in at3624
\new odd
\in at4730
\new odd
\in at5070
\new odd
\in at5100
\new odd
\in at910
\new odd
\in at912
\new odd
\in atatmcut
\new odd
\in atauth
\new odd
\in atthisf
\new odd
\in athiso
\new odd
\in atfuj
\new odd
\in atfuj7
\new odd
\in atomron
\new odd
```

```
\in atsetl
\new odd
\n attidl
\new odd
\n attrit
\new odd
\n atutils
```

- b. For example purposes, assume that you only want to print event messages for the AT1000, ATAUTH, and ATSETL PMIDs. In this case, you would delete the files from XPRINTAT that you do not want to be printed. The modified XPRINTAT file would then look as follows. You can also create your own customized print files by copying existing print files and modifying them.

```
\new odd
\n at1000
\new odd
\n atauth
\new odd
\n atsetl
```

6. Exit the HP NonStop editor when finished to print the files.
7. Enter the appropriate TGAL commands from the TACL prompt to print the files you want. Use the instructions provided in the “Printing Event Message Documentation” topic later in this section to print the appropriate event message documentation files.
8. Assemble all of your printed event message documentation (perhaps on fanfold paper) in alphabetical order, by PMID, and place it in a binder. The printed event messages can then be kept in a central location so you can use them as needed.

Note: The page numbering scheme of the event message documentation contains the PMID and page number. This format assists you in alphabetizing your printed event message documentation by PMID. For more information on file page layout, refer to section 2.

Locating Event Message Documentation

The following instructions describe how to locate event message documentation on the HP NonStop computer:

1. Identify the PMID associated with the product module that generated the event message. The originating source and PMID of each message can be determined from the message as it is displayed on the log.
2. Log on to the HP NonStop using your assigned logon and password. After successful logon, a TACL prompt is displayed.
3. Enter the VOLUME command from the TACL prompt to move to the HP NonStop subvolume where the event message documentation is located. If you do not know the subvolume where the documentation files are located, refer to section 2 for a complete list of PMIDs and the subvolumes on which they reside. In this example, assume the documentation is located on the BA60LOGM subvolume.

```
VOLUME $REL60.BA60LOGM
```

4. List the files on this subvolume by entering the following command from the TACL prompt.

```
FILES
```

BASE24 event messages are documented on this subvolume according to the product module that generated each message and are organized alphabetically by product module IDs (PMIDs). For example, all of the event messages generated by the BASE24-pos Settlement product module are documented in the PSSETL file.

The following is an example of the output displayed when this command is entered.

```
$REL60.BA60LOGM
```

```
AT1000    AT24V7    AT3624    AT4730    AT5070    AT910     AT912     ATATMCUT
ATAUTH    ATFUJ     ATFUJ7    ATHISF    ATHISO    ATSETL    ATTIDL    ATTRIT
ATUTILS   BABAMCUT  BABAUTIL  BAHISF    BAHISO    BAREFR    BASUPERX  BATAR
FHM       FHMHISF   FHMHISO   PSAUTH    PSCRTA    PSHISF    PSHISO    PSHYPRCM
PSMHI     PSMTTL    PSNCRNDP  PSPSMCUT  PSROUTER  PSSETL    PSSPDH    PSV1DH
PSV2DH    README   SECUTILS  TBHISO    TLRHISO   TRAUTH    TRSETL    XPRINTAT
XPRINTBA  XPRINTPS  XPRINTTR
```

-
5. Locate the file on the subvolume by the product module ID (PMID) corresponding to the product module for which you want event message documentation. If you do not know the PMID, refer to section 2 for a complete list of PMIDs and their corresponding product module names.
 6. Access the event message documentation for the specified product module by using a HP NonStop editor. For more information on HP NonStop text editors, refer to the appropriate HP NonStop manual.
 7. Search for the event message based on how the event messages for the product module are documented. Typically, the event messages are documented by event message number. However, some product modules do not generate event message numbers. In these cases, you can search for event messages using a text string.
 8. Exit the file when finished.

Accessing Special Use Files

In addition to the event message files that reside on the various HP NonStop subvolumes, there are several special use files. These files are edit files that you can access using a text editor. Special use files provide information about the event message documentation or commands to print the event message files for a specific product.

The README file on the BA xx LOGM subvolume, where xx is the number of the current release, contains information on the following topics:

- Subvolume content and organization
- Accessing event message documentation
- Event message documentation file content
- Instructions for printing single and multiple files

The XPRINT xx files contain the TGAL commands required to print out all the event message documentation for a particular product, where xx identifies the product. The valid product identifiers for these files are listed in section 2.

Printing Event Message Documentation

You can choose from several options available to print event messages in hard copy form. You can print an individual file or multiple files through use of the HP NonStop TGAL utility. TGAL allows you to print an entire file, a specific page from a file, or a range of pages from a file. This is done by simply altering the TGAL commands to achieve the end result you want. For more information on the TGAL utility, refer to the HP NonStop *TGAL Manual*.

This subsection describes some of the options most commonly used for printing event message documentation. Each example includes an explanation of the print option, the TGAL command you enter to use the option, and the results of entering the TGAL command.

Printing a Single File

You can print an individual file from the HP NonStop subvolume by entering TGAL commands from a TACL prompt. The following paragraphs describe in detail the print options you can select when printing a single file.

Printing the Entire File

To print an entire file, enter the following TGAL command from a TACL prompt:

Syntax

TGAL/IN *file-name*, OUT *spooler-location*/

IN *file-name* — The name of the event message documentation file to be printed.

OUT *spooler-location* — The HP NonStop spooler location where the formatted event message documentation is printed.

Example

TGAL/IN AT1000, OUT \$S.#HOLD/

Upon entry of this command, the TGAL formatted documentation for the BASE24-atm Diebold 10XX/478X Device Handler process is sent to the #HOLD spooler location.

Printing a Specified Range of Pages

If you only require certain pages be printed from a specific file, you can indicate a consecutive range of pages you want and print only those pages. Page ranges are printed consecutively. For example, if you specify to print out pages 10 through 15, pages 10, 11, 12, 13, 14, and 15 are printed. If you want to print certain page ranges that have intervening pages that you do not want to be printed (e.g., 10, 11, 14, 17) you must send multiple TGAL commands.

You can print a specific range of pages from a file by entering the OUT command and the designated range of pages immediately following the second slash (/) in the TGAL command. A slash is also used to separate the range of pages. To use this option, enter the following TGAL command from a TACL prompt.

Syntax

TGAL/IN *file-name*, OUT *spooler-location*/OUT *n1/n2*

IN *file-name* — The name of the event message documentation file to be printed.

OUT *spooler-location* — The HP NonStop spooler location where the specified formatted event message documentation pages are printed.

OUT *n1/n2* — The range of pages to be printed, where *n1* is the number of the first page in the range and *n2* is the number of the last page in the range.

Example

TGAL/IN AT1000,OUT \$S.#HOLD/OUT 10/15

Upon entry of this command, the formatted pages 10 through 15 of the BASE24-atm Diebold 10XX/478X Device Handler process event message documentation are sent to the designated spooler location.

Printing a Single Page

You can also print a single page from a file by entering the OUT command and the designated page immediately following the second slash (/) in the TGAL command. After entering the OUT command, you must designate the number of the page you want to print by entering the page number followed by a slash and then entering the page number again (e.g., 9/9, 10/10, 22/22, etc.). This indicates that only one page should be printed. To use this option, enter the following TGAL command from a TACL prompt.

Syntax

TGAL/IN *file-name*, OUT *spooler-location*/OUT *n/n*

IN *file-name* — The name of the event message documentation file to be printed.

OUT *spooler-location* — The HP NonStop spooler location where the specified formatted event message documentation page is printed.

OUT *n/n* — The range of pages to be printed, where *n* is the number of a single page to be printed.

Example

TGAL/IN AT1000,OUT \$S.#HOLD/OUT 10/10

Upon entry of this command, the formatted page 10 of the BASE24-atm Diebold 10XX/478X Device Handler process event message documentation is sent to the designated spooler location.

Printing Multiple Files

You can also print multiple files at one time from the HP NonStop subvolume. For example, if you require an entire set of event message documentation for a specific BASE24 product (e.g., BASE24-atm, BASE24-pos, or BASE24-teller), you can use the appropriate XPRINT_{xx} file, where *xx* identifies a particular BASE24 product. The syntax for using print files is shown below.

Syntax

TGAL/IN *print-file-name*, OUT *spooler-location*/

IN *print-file-name* — the name of the event message documentation print file to be executed.

OUT *spooler-location* — the HP NonStop spooler location where the specified formatted event message documentation is printed.

Example

TGAL/IN XPRINTAT,OUT \$S.#HOLD

Upon entry of this command, all of the event message documentation for the BASE24-atm product is printed. The documentation for each product module included begins on a new page.

Converting TGAL Files into TFORM

Along with TGAL, HP NonStop supports a newer text formatter called PS Text Format (TFORM). If you are more familiar with TFORM, you can convert your TGAL files to TFORM files. This conversion allows you to use TFORM instead of TGAL when formatting files for printing. Enter the following command at a TACL prompt to convert your TGAL files to TFORM:

Syntax

TFORMCVT/IN *tgal-file-name*, OUT *tform-file-name*/ TGAL

IN *tgal-file-name* — The name of the TGAL event message documentation file to be converted.

OUT *tform-file-name* — The TFORM event message documentation file to be created.

Example

TFORMCVT/IN AT1000, OUT AT1000TF/ TGAL

Upon entry of this command, a new event message documentation file named AT1000TF is created with TFORM formatting commands. The existing AT1000 TGAL file is not changed.

Printing Files Using TFORM

Files can also be printed using TFORM. Printing files using TFORM is similar to printing files using TGAL. To print a TFORM file, enter the following command from a TACL prompt:

To print an entire file, enter the following TFORM command from a TACL prompt:

Syntax

TFORM/IN *file-name*, OUT *spooler-location*/

IN *file-name* — The name of the event message documentation file to be printed.

OUT *spooler-location* — The HP NonStop spooler location where the formatted event message documentation is printed.

Example

TFORMCVT/IN AT1000TF, OUT \$S.#HOLD

Upon entry of this command, the TFORM formatted documentation for the BASE24-atm Diebold 10XX/478X Device Handler process is sent to the #HOLD spooler location.

For more information on TFORM, refer to the HP NonStop ***PS TEXT FORMAT Reference Manual***.

Accessing Event Message Documentation from the CD-ROM

All BASE24 event message documentation files on the BAxxLOGM and OKxxLOGM subvolumes, where xx is the number of the current release, and switch-specific subvolumes are also available on the BASE24 Product Documentation CD-ROM. Event message files are listed by PMID and process name on the BASE24 Event Message Documentation Menu. You can search for documentation on a particular message in two ways. You can search for the documentation in all event message files by loading the event message documentation indexes and searching on the PMID and event number and using the Search function. You can also open an individual event message documentation file and use the Find function to search for the event number.

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4: The Manual Generator Utility

The Manual Generator utility allows you to automatically generate event message and TAL library documentation files from EMS templates and TAL library source files, respectively.

This section describes the command syntax used to run the Manual Generator utility, and provides procedures and examples for generating event message and TAL library source documentation. It also describes the standard formats EMS template and TAL source files must follow for the Manual Generator utility to work properly and how title pages can be used with the utility.

Generating Event Message Documentation

The Manual Generator utility allows you to generate event message documentation for EMS template files. Event message documentation should be generated any time after you modify the text, cause, or remedy for one or more event messages in an EMS template source file. ACI provides EMS template files for all object-oriented products.

When you modify the text of an event message, the change does not take effect until after you have performed several tasks to incorporate the modified templates into the system templates. The procedures for incorporating modified EMS templates in object-oriented products are provided in the ACI informal ***Object-Oriented Programming Standards Manual***.

The Manual Generator utility works with any EMS template source file that uses commented flower boxes that adhere to BASE24 release 5.0 standards. A description of these standards is provided at the end of this section along with an example flower box and the corresponding output generated by the utility.

The event message documentation output generated by the utility adheres to the page layout standards described in section 2. The generated event message documentation output files include TGAL formatting commands. These files can be printed using the procedures provided in section 3.

Generating TAL Library Documentation

Besides event message documentation, the Manual Generator utility allows you to generate TAL library documentation for TAL library source files. ACI provides a set of standard core and kernel TAL libraries for all object-oriented products. TAL library documentation is intended for anyone using the ACI object-oriented environment to develop software. For more information on TAL libraries, refer to the informal *Object-Oriented Programming Standards Manual*.

The Manual Generator utility works with any TAL library source file that uses commented flower boxes that adhere to ACI BASE24 release 5.0 programming standards. A description of these standards is provided at the end of this section along with an example flower box and the corresponding output generated by the utility.

The TAL library documentation output generated by the utility also adheres to the page layout standards described in section 2, except that page numbers are prefaced by the “LIB” characters. The generated TAL library documentation output files include TGAL formatting commands. These files can be printed using the procedures provided in section 3.

Running the Manual Generator Utility

The Manual Generator utility can be run noninteractively using an obey file or interactively from a TACL prompt. Both methods of running the utility are described below and on the following pages.

Run options that can be set from a screen prompt when the utility is run interactively are fully explained in the run command syntax used when running the utility noninteractively from an obey file.

Note: If additional EMS template files are *sourced in* to a specified EMS template input file, either the defines for the sourced in files must be resolved (e.g., by mapping the defines) prior to running the utility (interactively or noninteractively) or the NOSOURCE option should be used. Sourced in files are identified by the ?SOURCE compiler directive in EMS template source files. For more information on mapping defines, refer to the HP NonStop *TACL Reference Manual* or *Guardian User's Guide*.

Running the Manual Generator Utility Noninteractively

The command syntax for running the Manual Generator utility noninteractively from within an obey file is described below. Items enclosed in square brackets ([]) are optional. Items not enclosed in square brackets are required and must be entered exactly as they appear. Italicized items are variables and are defined in the paragraphs that follow.

Syntax

ASSIGN WORK-FILE, *work-file-location*
[ASSIGN TOC-FILE, *toc-file-location*]
[ASSIGN TITLEPAGE-FILE, *titlepage-file-location*]
[RUN] [\node.\$volume.subvolume.] MANGENTO [/ [IN *source-file*]
[, OUT *\$spooler*] /] [[*option*] [, *option*] ...]

ASSIGN WORK-FILE *work-file-location* — The name of the documentation output file to be generated. If a node, volume, and subvolume are not included in the file name, the file is generated on the node, volume, and subvolume where the Manual Generator utility object code resides.

ASSIGN TOC-FILE *toc-file-location* — The name of the table of contents output file to be generated when generating TAL library documentation. If a node, volume, and subvolume are not included in the file name, the file is generated on the node, volume, and subvolume where the Manual Generator utility object code resides.

ASSIGN TITLEPAGE-FILE *titlepage-file-location* — The name of the title page file to be specified on the first line of the documentation output file. If a node, volume, and subvolume are not included in the file name, the node, volume, and subvolume where the Manual Generator utility object code resides are used. If this assign is not specified, it defaults to the MGENITL file on the node, volume, and subvolume where the utility object file resides.

RUN — An optional keyword for the RUN command. If this keyword is included in the command, either the Manual Generator utility object code file must be located on the current node, volume, and subvolume or the location must be included in the command. If this keyword is not included in the command, the Manual Generator utility must be located on a volume and subvolume that is included in the PMSEARCHLIST for your TACL logon.

\node.volume.subvolume. — Specifies the location of the Manual Generator utility. This variable, which can include the node, volume, and subvolume, is required if the utility is not on the current node, volume, or subvolume and the location of the utility is not included in your PMSEARCHLIST. If this variable is included in the command, the keyword RUN must also be included in the command.

IN *source-file* — The name of the TAL library source file or EMS template source file to be used as input to the Manual Generator utility. If a node, volume, and subvolume are not included in the file name, the file must reside on the same node, volume, and subvolume as the Manual Generator utility object code file.

OUT *\$spooler* — The location of the spooler report generated when the Manual Generator utility is run noninteractively. This report summarizes the processing performed by the utility.

option — A Manual Generator run option defining the processing performed and output required from the utility. Multiple options can be entered to specify and control the processing and output desired. Each of these options is described on the following pages.

Example

```
ASSIGN WORK-FILE, \PRO1.$DATA.OK60LOGM.ISKNANS  
  RUN \PRO1.$DATA.ACILIBI.MANGENTO/IN \PRO1.$DATA.&  
  OK60ANS.ANSPS,OUT $$.#MGEN/STANDARD,&  
  SAVEWORK,SOURCE,TGAL,NOTITLE
```

This example generates event message documentation for the Application Network Services object. The work file is generated in the OK60LOGM subvolume in standard format without a title page and is saved. Additional EMS template files are sourced in and a formatted copy of the documentation is sent to the spooler.

Note: In the above example, defines for sourced in template files must be mapped prior to running the obey file.

Example

```
ASSIGN WORK-FILE, \PRO1.$DATA.OK60LIB.OKLIBMAN  
  ASSIGN TOC-FILE, \PRO1.$DATA.OK60LIB.OKLIBTOC  
  ASSIGN TITLEPAGE-FILE, \PRO1.$DATA.ACILIBI.MGENACI  
  RUN \PRO1.$DATA.ACILIBI.MANGENTO/IN \PRO1.$DATA.&  
  OK60LIB.OKLIBTS,OUT $$.#MGEN/STANDARD,&  
  SAVEWORK,SAVETOC,TGAL,TITLE,FORCEODD,
```

This example generates TAL library documentation for kernel libraries. The work file and table of contents file are generated in the OK60LIB subvolume in standard format and are saved. A blank page is automatically inserted between the title page and table of contents to force the table of contents to begin on an odd-numbered page. If the generated table of contents ends on an odd-numbered page, another blank page is automatically inserted between the table of contents and the library documentation, forcing the documentation to begin on an odd-numbered page. A formatted copy of the documentation using an ACI title page is sent to the spooler.

Sample Obey File

The following is a sample obey file named GOMGEN that generates event message documentation for the SQL Refresh process. In this example, a title page is not used, the documentation is not sent to the spooler, source statements are ignored, and the work file is saved.

```
clear all assign
assign work-file, $data.ok60logm.refrps

run $data.acilibi.mangento /in $data.oc60rfsq.refrps &
    , out $s.#mgen/ &
    notitle,notgal,nosource,savework
```

Manual Generator Run Options

The following Manual Generator run options control the processing performed by the utility or the appearance of the documentation output when you run the utility noninteractively. Options can be entered on the run command line in any order.

Option	Description
FORCEODD NOFORCEODD	Indicates whether the table of contents and the first page of the TAL library documentation following the table of contents are forced to begin on an odd page. If the FORCEODD and TITLE options are set, the utility inserts an “intentionally left blank” page after the title page. If the FORCEODD option is set and the generated table of contents ends on an odd-numbered page, the utility inserts another “intentionally left blank” page after the table of contents to force the first page of documentation to an odd-numbered page. If the NOFORCEODD option is set, no blank pages are inserted. The FORCEODD option is canceled by the NOFORCEODD option, and vice versa. Default: NOFORCEODD

Option	Description
FORMATTED STANDARD	Indicates whether documentation output files include additional TGAL formatting commands such as \need, \ov, and \center to enhance the appearance of printed output. If the FORMATTED option is set, these additional commands are added. If the STANDARD option is set, these additional commands are not added. The FORMATTED option is canceled by the STANDARD option, and vice versa. Note: Because the FORMATTED option introduces additional formatting commands that can make documentation files harder to read online, ACI recommends that this option not be used when generating documentation to be viewed online. Default: STANDARD
SAVETOC NOSAVETOC	When generating TAL library documentation, this option indicates whether the table of contents file is saved or purged when processing is completed. If the SAVETOC option is set, the table of contents file is saved. If the NOSAVETOC option is set, the table of contents file is purged when processing is completed. The SAVETOC option is canceled by the NOSAVETOC option, and vice versa. When you save the table of contents file, you can subsequently reprint the table of contents using TGAL without having to regenerate the file. This option has no impact when generating event message documentation. Default: SAVETOC

Option	Description
SAVEWORK NOSAVEWORK	Specifies whether the output documentation file (work file) is saved or purged once processing is completed. If the SAVEWORK option is set, the work file is saved. If the NOSAVEWORK option is set, the work file is purged. The SAVEWORK option is canceled by the NOSAVEWORK option, and vice versa. When you save the work file, you can subsequently reprint the documentation using TGAL without having to regenerate the work file. The NOSAVEWORK option should only be used if the TGAL option is set. The TGAL option causes the contents of the work file to be sent to a spooler location before the file is purged. Default: SAVEWORK
SOURCE NOSOURCE	Indicates whether additional EMS template source files identified in the specified EMS template source file are sourced in. If this option is set to SOURCE, the additional template files are sourced in. If this option is set to NOSOURCE, additional template files are not sourced in. The SOURCE option is canceled by the NOSOURCE option, and vice versa. Additional EMS template source files are identified in ?SOURCE statements within an EMS template source file. If these ?SOURCE statements use defines to locate the additional files, the defines must be mapped prior to running the utility. This option has no impact when generating TAL library documentation. Default: SOURCE

Option	Description
TGAL NOTGAL	Specifies whether the Manual Generator utility should start a TGAL process to process the work file and create a formatted spooler document. If this option is set to TGAL, a TGAL process is started to process the work file. If this option is set to NOTGAL, a TGAL process is not started. The TGAL option is canceled by the NOTGAL option, and vice versa. If the TGAL option is set, the process generates the formatted document in the <code>\$\$.<src-file-minus-2></code> spooler location, where <code><src-file-minus-2></code> is the name of the source file processed by the utility minus the last two characters of the file name. For example, if the source file is ANSPS, the formatted output would be placed in the <code>\$\$.#ANS</code> spooler location. Default: TGAL
TITLE NOTITLE	Indicates whether a title page is included in the output file. If this option is set to TITLE, a TGAL \in command line specifying a title page file is added to the first line of the output file. If this option is set to NOTITLE, this command line is not added to the output file. The TITLE option is canceled by the NOTITLE option, and vice versa. When the TITLE option is set, the utility uses the TITLEPAGE-FILE assign. If this assign is not specified, it defaults to the MGENTITL file on the same node, volume, and subvolume as the utility object file. Default: TITLE

Option	Description
TOC NOTOC	When generating TAL library documentation, this option indicates whether a table of contents is generated. If this option is set to TOC, a table of contents is generated. If this option is set to NOTOC, a table of contents is not generated. The TOC option is canceled by the NOTOC option, and vice versa. When the TOC option is set, the utility requires the TOC-FILE assign. There is no default for this assign. If the file name is not fully qualified, the node, volume, and subvolume on which the utility object file resides are presumed. This option has no impact when generating event message documentation. Default: TOC

Sample Spooler Report

A report summarizing processing is sent to the spooler each time you run the Manual Generator utility noninteractively from an obey file. This report lists the run options selected for the run, the files used or created, and the processing performed. The report is generated in the spooler location specified in the run command. A sample spooler processing report is shown below. In this example, the utility encountered four ?SOURCE statements in the input file that were not used because the NOSOURCE option was set.

```
Run Parameters: TITLE, TOC, TGAL, SAVEWORK,
                SAVETOC, STANDARD, NOFORCEODD

Input File:  $DATA.OK60ANS.ANSPS

New Work File created: $data.ok60logm.isknans
Processing $DATA.OK60ANS.ANSPS
Run parameter, NOSOURCE is in effect.  Source statements will not be followed.
Run parameter, NOSOURCE is in effect.  Source statements will not be followed.
Run parameter, NOSOURCE is in effect.  Source statements will not be followed.
Run parameter, NOSOURCE is in effect.  Source statements will not be followed.

TGAL in progress

Processing complete
```

Event Message Documentation File Considerations

When using the Manual Generator utility to generate an online event message documentation file, the following run options should be in force: NOTITLE, SAVEWORK, SOURCE, and STANDARD. To enhance the online readability of the documentation, you should use the STANDARD layout to minimize the TGAL commands added to the file.

After the event message documentation file is generated, you may still need to manually edit the file to remove excessive \need on and \need off commands.

Generating Event Message Documentation Interactively

When you run the Manual Generator utility interactively from a TACL prompt, the utility prompts you for all needed information. You provide the information required at each prompt by typing it. At prompts that request the name of a file, you must also press the **Enter** key to enter the information. At prompts that request a single character value, the value is entered as soon as you type it, without pressing the **Enter** key. You can exit the utility at any time by pressing the **Break** key or **Control-Y** keys.

Some prompts time out if you do not enter a value within 10 seconds after the prompt is displayed.

1. At the TACL prompt, enter the following command to start the Manual Generator utility, where [*\$vol.subvol*] identifies the volume and subvolume location of the utility if it is not on the current volume and subvolume.

`[ $vol.subvol. ]MANGENTO`
2. Enter the input file name (source file) or the value H to display the Help menu. If the input file is not on the current volume and subvolume, the file name must be qualified. After you enter the input file name, the fully qualified input file name is displayed near the top of your screen—"Input File: *\$vol.subvol.input-file-name*." The utility opens and reads the specified input file to determine whether it is an EMS template source file or a TAL library source file. The remaining prompts displayed are tailored to the type of file found.

If you enter the value H at this prompt, refer to the "Displaying Help Information" discussion for the prompts displayed while the help facility is in use.

Prompt: Enter Input file name (H for Help Menu):

Entry: *input-file-name* or H

3. Enter the output file name. If you do not qualify the file name with a volume and subvolume location, it defaults to the volume and subvolume where the Manual Generator utility object code is located. The fully qualified output file name is displayed near the top of your screen—"Output File: *\$vol.subvol.output-file-name*."

Prompt: Enter Output file name:

Entry: *output-file-name*

-
4. If the specified output file already exists, the utility prompts you to indicate whether the existing file should be purged. If the specified output file does not already exist, this prompt is not displayed. If the prompt times out or you press the **Enter** key without specifying a value, a value of Y is assumed.

Prompt: *\$vol.subvol.output-file-name* already exists. Shall I PURGE (N/Y)?

Entry: N or Y

5. Indicate whether you want a title page to be included in the documentation. If the prompt times out, it defaults to a value of Y. If you press the **Enter** key without entering a value, a value of N is assumed.

Prompt: Do you require a Title Page (N/Y)?

Entry: N or Y

6. If you entered the value Y at the previous prompt, enter the file name for the title page to be used for the documentation. If the prompt times out or you press the **Enter** key without specifying the title page, the default title page (i.e., MGENTITL) is used. If you entered the value N at the previous prompt, this prompt is not displayed.

Prompt: Enter Title Page file name (Enter for default):

Entry: *title-page-file-name*

7. Specify the output format for the documentation. If the prompt times out or you press the **Enter** key without specifying the format, the default output format (i.e., formatted) is used.

Prompt: Which Output format do you prefer; Formatted or Standard (F/S)?

Entry: F or S

At this point, the utility begins processing the source file. Informational messages are displayed on the screen as processing progresses.

8. When the utility finishes processing the source file, the utility prompts you as to whether the utility should start a TGAL process to generate formatted output to the spooler. If the prompt times out or you press the **Enter** key without entering a value, the prompt defaults to a value of Y (i.e., a TGAL process is started). If you select a TGAL process to be started, the fully qualified name of a temporary TGAL output file is displayed near the top of your screen—"Temporary TGAL Output File: *\$vol.subvol.tgalfile*." The utility uses the name of the input file to name the temporary TGAL output file. The utility drops the last two character from the input file name and appends the ZZ characters to the file name. After processing, this temporary file is automatically purged by the utility.

If you enter the value Y at this prompt, the TGAL process begins processing the temporary file. Informational messages are displayed on the screen as processing progresses. The formatted output is generated at the `$$.#infile` spooler location, where *infile* is the name of the input file minus the last two characters.

The message “TGAL in progress” is displayed while the TGAL process executes the output file.

Prompt: Shall I invoke TGAL to process `$vol.subvol.output-file-name` (N/Y)?

Entry: N or Y

9. Specify whether the utility should purge the work file. If the prompt times out or you press the **Enter** key without entering a value, the prompt defaults to a value of N (i.e., the work file is saved).

Prompt: Shall I Purge the Work File (N/Y)?

Entry: N or Y

When the utility completes processing, the message “Processing Complete” is displayed on the screen and the TACL prompt is redisplayed. The event message documentation is contained in the specified work file or spooler location.

Generating TAL Library Documentation Interactively

When you run the Manual Generator utility interactively from a TACL prompt to generate TAL library documentation, the utility prompts you for all needed information. You provide the information required at each prompt by typing it. At prompts that request the name of a file, you must also press the **Enter** key to enter the information. At prompts that request a single character value, the value is entered as soon as you type it, without pressing the **Enter** key. You can exit the utility at any time by pressing the **Break** key or **Control-Y** keys.

Some prompts time out if you do not enter a value within 10 seconds after the prompt is displayed.

1. At the TACL prompt, enter the following command to start the Manual Generator utility, where `[$vol.subvol]` identifies the volume and subvolume location of the utility if it is not on the current volume and subvolume.

```
[ $vol . subvol . ] MANGENTO
```

-
2. Enter the input file name or the value H to display the Help menu. If the input file is not on the current volume and subvolume, the file name must be qualified. The utility opens and reads the specified input file to determine whether it is an EMS template source file or a TAL library source file. The remaining prompts displayed are tailored to the type of file found. The fully qualified input file name is displayed near the top of your screen—"Input File: *\$vol.subvol.input-file-name*."

If you enter the value H at this prompt, refer to the "Displaying Help Information" discussion later in this section for the prompts displayed while the help facility is in use.

Prompt: Enter Input file name (H for Help):

Entry: *input-file-name* or H

3. Enter the output file name. If you do not qualify the file name with a volume and subvolume location, it defaults to the volume and subvolume where the Manual Generator utility object code is located. After you enter the output file name, the fully qualified output file name is displayed near the top of your screen—"Output File: *\$vol.subvol.output-file-name*."

Prompt: Enter Output file name:

Entry: *output-file-name*

4. If the specified output file already exists, the utility prompts you to indicate whether the existing file should be purged. If the specified output file does not already exist, this prompt is not displayed. If the prompt times out or you press the **Enter** key without specifying a value, a value of Y is assumed.

Prompt: *\$vol.subvol.output-file-name* already exists. Shall I PURGE (N/Y)?

Entry: N or Y

5. Indicate whether you want a title page to be included in the documentation. If the prompt times out, it defaults to a value of Y. If you press the **Enter** key without entering a value, a value of N is assumed.

Prompt: Do you require a Title Page (N/Y)?

Entry: N or Y

6. If you entered the value Y at the previous prompt, enter the file name for the title page to be used for the documentation. If the prompt times out or you press the **Enter** key without specifying the title page, the default title page (i.e., MGENTITL) is used. If you entered the value N at the previous prompt, this prompt is not displayed.

Prompt: Enter Title Page file name (Enter for default):

Entry: *title-page-file-name*

7. Specify the output format for the documentation. If the prompt times out or you press the **Enter** key without specifying the format, the default output format (i.e., formatted) is used.

Prompt: Which Output format do you prefer; Formatted or Standard (F/S)?

Entry: F or S

8. Indicate whether you want the utility to generate a table of contents for the documentation. If the prompt times out or you press the **Enter** key without entering a value, the prompt defaults to a value of Y (i.e., a table of contents is generated).

If you enter the value N at this prompt, the utility begins processing the source file and you can continue with step 11. Informational messages are displayed on the screen as processing progresses.

If you enter the value Y at this prompt, proceed with step 9.

Prompt: Do you require a Table of Contents (N/Y)?

Entry: N or Y

9. If you entered the value Y at the previous prompt, enter the file name of the table of contents file to be generated. If you press the **Enter** key without entering a file name, the utility uses the name of the input file to name the table of contents file. The utility drops the last two character from the input file name and appends the ZX characters to the file name. The fully qualified name of the temporary table of contents file is displayed near the top of your screen—"Table of Contents File: *\$vol.subvol.tocfile*."

Prompt: Enter TOC file name (ENTER for default):

Entry: *toc-file-name*

10. If you selected a table of contents to be generated, indicate whether you want the table of contents and the TAL library documentation to begin on odd-numbered pages. If this option is selected and a title page is to be included, the utility inserts a blank page between the title page and the table of contents. If this option is selected and the table of contents ends on an odd page, the utility inserts another blank page at the end of the table of contents to force the TAL library documentation to also begin on an odd-numbered page. If the prompt times out or you press the **Enter** key without entering a value, the prompt defaults to a value of N (i.e., a blank page is not inserted if the table of contents ends on an odd-numbered page).

Prompt: Do you require "Force Odd" paging for double-sided printing (N/Y)?

Entry: N or Y

At this point, the utility begins processing the source file. Informational messages are displayed on the screen as processing progresses.

11. When the utility finishes processing the source file, the utility prompts you as to whether the utility should start a TGAL process to generate formatted output to the spooler. If the prompt times out or you press the **Enter** key without entering a value, the prompt defaults to a value of Y (i.e., a TGAL process is started). If you select a TGAL process to be started, the fully qualified name of a temporary TGAL output file is displayed near the top of your screen—"Temporary TGAL Output File: *\$vol.subvol.tgalfile*." The utility uses the name of the input file to name the temporary TGAL output file. The utility drops the last two character from the input file name and appends the ZZ characters to the file name. After processing, this temporary file is automatically purged by the utility.

If you enter the value Y at this prompt, the TGAL process begins processing the temporary file. Informational messages are displayed on the screen as processing progresses. The formatted output is generated at the *\$.#infile* spooler location, where *infile* is the name of the input file minus the last two characters.

The message "TGAL in progress" is displayed while the TGAL process executes the output file.

Prompt: Shall I invoke TGAL to process *\$vol.subvol.output-file-name* (N/Y)?

Entry: N or Y

12. Specify whether the utility should purge the table of contents file. If the prompt times out or you press the **Enter** key without entering a value, the prompt defaults to a value of N (i.e., do not purge the table of contents file).

Prompt: Shall I Purge the temporary Table of Contents File (N/Y)?

Entry: N or Y

13. Specify whether the utility should purge the work file. If the prompt times out or you press the **Enter** key without entering a value, the prompt defaults to a value of N (i.e., the work file is saved).

Prompt: Shall I Purge the Work File (N/Y)?

Entry: N or Y

At this point, the message "Processing Complete" is displayed on the screen and the TACL prompt is redisplayed. The TAL library documentation is contained in the specified work file or spooler location. The table of contents is contained in the specified temporary table of contents file or spooler location.

Displaying Help Information

When you run the Manual Generator utility interactively, you can access online help information. To access help information, perform the following steps.

1. At the TACL prompt, enter the following command to start the Manual Generator utility, where [*\$vol.subvol*] identifies the volume and subvolume location of the utility if it is not on the current volume and subvolume.

[*\$vol.subvol.*]MANGENTO

2. Enter the value H to display the Help menu.

Prompt: Enter Input file name (H for Help Menu):

Entry: H

3. From the Help menu, specify whether you want to view a general description of the Manual Generator utility, command syntax for running the utility noninteractively, or flower box examples. You can also exit the help facility or exit the utility completely (i.e., redisplay the TACL prompt) from this menu.

If you type the value 1 at this prompt, continue with step 4. If you type the value 2 at this prompt, skip to step 5. If you type the value 3 at this prompt, skip to step 6.

Prompt: Choices: (1) General Description
(2) Running MANGEN non-interactively
(3) Release 5.0 Flower Box Examples
(4) Exit Help
(5) Abort Program

Choice (1 - 5)?

Entry: 1-5

4. If you typed the value 1 at the previous prompt, a general description of the Manual Generator utility is displayed. After reading this description, you can specify whether you want to continue viewing help information or exit the utility. If you type the value N at this prompt, the Manual Generator utility is stopped and the TACL prompt is redisplayed. If you type the value Y at this prompt, the Help menu is redisplayed.

Prompt: Do you wish to continue (N/Y)?

Entry: N or Y

-
5. If you typed the value 2 at the prompt in step 3, the command syntax for running the Manual Generator utility noninteractively is displayed. After reading this information, you can specify whether you want to continue viewing help information or exit the utility. If you type the value N at this prompt, the Manual Generator utility is stopped and the TACL prompt is redisplayed. If you type the value Y at this prompt, the Help menu is redisplayed.

Prompt: Do you wish to continue (N/Y)?

Entry: N or Y

6. If you typed 3 at the prompt in step 3, the Flower Box Help menu is displayed. From this menu, you can select to view a TAL library or EMS template flower box example, redisplay the Help Menu (Exit Help), or exit the utility completely (i.e., redisplay the TACL prompt).

If you type a value of 1 or 2 at this prompt, continue with step 7.

Prompt: Choices: (1) Display TAL Library Flower Box
(2) Display EMS Template Flower Box
(3) Exit Help
(4) Abort Program

Choice (1 - 4)?

Entry: 1-4

7. If you typed a value of 1 or 2 at the Flower Box Help menu prompt, the appropriate flower box example is displayed. A prompt at the top of the screen then allows you to view the first page of the flower box example, view the next page of the flower box example, return to the flower box help menu, or exit the utility completely.

Prompt: Choices: (1) 1st pg, (2) Next Pg, (3) Return, (4) Abort
Prog. (1 - 4)?

Entry: 1-4

EMS Template Source File Standards

For the Manual Generator utility to work properly, all introductory, file history and event message documentation must be provided in specially named files and in specific formats. These file and format requirements are discussed below.

Files

All event message documentation for a specific subsystem is provided in two files. Event message introductory text for a subsystem is provided in an `xxxxxxAS` file, where `xxxxxx` is a 1- to 6-character identifier for the subsystem. Individual event message documentation for a subsystem, including a history of changes, is provided in an EMS source template file named `xxxxxxPS`, where `xxxxxx` is a 1- to 6-character identifier for the subsystem. The 1- to 6-character subsystem identifiers must be exactly the same in both files in the same subsystem and both files must reside on the same HP NonStop subvolume when the Manual Generator utility is run.

The introductory text file for a subsystem is an edit file with TGAL commands that set the section name, head, and subhead for an event message documentation file. The file also specifies the title, product module ID (PMID) for the subsystem, any hard-coded generator IDs for the subsystem, and copyright information.

The EMS template source file for a subsystem contains HP NonStop Template-Language statements for each event, file history comments, and event message documentation comments for each event message template. The format of the HP NonStop Template-Language statements used by ACI is described in the *NET24-XPNET Event Management Manual*. The formats required for file history and event message documentation comments are described below. The Template-Language statements are not used by the Manual Generator utility.

Introductory Text File

The introductory text file (e.g., `xxxxxxAS`) contains the introductory text for an event message documentation file. The Manual Generator utility places this text at the beginning of a generated event message documentation file. When run, the Manual Generator utility searches for the introductory text file on the same subvolume where the EMS template source file was found. The introductory text file for the Application Network Services object is shown on the next page.

```
\poff 7
\new
\sect "ISKNANS-"1
\head "BASE24 ITS APPLICATION NETWORK SERVICES (ANS) OBJECT"
\subhead "MODULE ID: ISKNANS"
```

```
-----
BASE24 ITS APPLICATION NETWORK SERVICES (ANS) OBJECT
EVENT MESSAGE DOCUMENTATION
-----
```

This file contains documentation for the event messages that can be generated by the BASE24 ITS ANS object. The product module ID (PMID) for the BASE24 ITS ANS object is ISKNANS. Event messages are documented in order by event message number.

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```
\new
```

EMS Template Source Files

File history and event message comments in EMS template source files provide the file history and event message documentation information in the generated event message documentation file. These comments must be provided in specific formats for use by the Manual Generator utility. All comments in EMS template

source files are denoted using the TACL comment characters—two equal signs (==). All characters following the two equal signs to the end of the line are considered to be comments.

The beginning of file history text is denoted by a comment line beginning with the ==HST characters. All subsequent comment lines, up to the characters “SSID” are included in the history section.

Individual event message documentation is provided in the EMS template source files in the form of commented *flower boxes*. A flower box is a contiguous series of comment lines in the shape of a box. All comment lines in the flower box for an event begin with either the ==EMS# or ==EMn# characters, where *n* is number from 1 to 4. Each number denotes specific event message information as shown in the following table.

Number	Denotes:
1	The beginning of event message documentation for an event. This comment line specifies the event number and is followed by subsequent comment lines that specify the message text, tokens, subject, and emphasis (severity).
2	The beginning of the probable cause for the event
3	The beginning of the action taken for the event.
4	The beginning of the recommended action for the event.

When the Manual Generator utility finds the “==EM1#” characters, it begins extracting information for an event. It continues to extract information for an event until it finds the “==EMS” characters followed by five pound sign (#) characters (#####). This signals the end of the documentation for an event.

Between the ==EM1# and ==EM2# comment lines, the following keywords signal specific information to the utility.

Keyword	Signals:
MESSAGE TEXT	The beginning of the event message text for an event.
TOKENS	The beginning of token descriptions for the event.
SUBJECT	The subject of tokens used in the event.
EMPHASIS	The severity of the event.

The Manual Generator utility requires that event message flower boxes adhere to the BASE24 release 5.0 flower box standards as depicted in the template below. The Manual Generator utility strips out all comment characters, EMS, EM n , and # characters, and all leading and trailing spaces for the generated output.

```
==EMS#####
==EMS#                                           #
==EM1#  00005:  EVT-LITERAL-GOES-HERE-L        #
==EMS#                                           #
==EMS#  MESSAGE TEXT:                          #
==EMS#    "MESSAGE TEXT - *TOKEN1 = <1>"        #
==EMS#                                           #
==EMS#  TOKENS:                                #
==EMS#    <1>:  Name:    IFMT-TKN-TOKEN1-L      #
==EMS#           Value:  IFMT-TNM-TOKEN1-L      #
==EMS#           Type:   ZSPI-TYP-TOKEN1        #
==EMS#           Descr:  TOKEN1 description     #
==EMS#                                           #
==EMS#  SUBJECT:  ACI-TKN-GENERATOR-L          #
==EMS#                                           #
==EMS#  EMPHASIS:  Critical                     #
==EMS#                                           #
==EMS#  ACTION ATTENTION EVENT:  No            #
==EMS#                                           #
==EM2#  PROBABLE CAUSE:                        #
==EMS#    Probable cause description goes here. #
==EMS#                                           #
==EM3#  ACTION TAKEN:                          #
==EMS#    Describe the action taken here.      #
==EMS#                                           #
==EM4#  RECOMMENDED ACTION:                    #
==EMS#    Describe the appropriate action here. Include any manual #
==EMS#    references.                          #
==EMS#                                           #
==EMS#####
```

Example EMS Template Source Flower Box

The following is an example of the flower box comments for an event in the EMS template source file for the Application Network Services object. The corresponding output generated by the Manual Generator utility for this event is shown on the next page.

```
==EMS#####==00054
==EMS#                                           ==00055
==EM1#  00010:  ANS-EVT-DEFINEPOOL-ERR-L        ==00056
==EMS#                                           ==00057
==EMS#  MESSAGE TEXT:                           ==00058
==EMS#    "DEFINEPOOL FAILURE - STATUS = <1>"    ==00059
==EMS#                                           ==00060
==EMS#  TOKENS:                                  ==00061
==EMS#    <1>:  Name:    ANS-TKN-STAT-L          ==00062
==EMS#           Value:  ANS-TNM-STAT-L          ==00063
==EMS#           Type:   ZSPI-TYP-INT            ==00064
==EMS#           Descr:  DEFINEPOOL status       ==00065
==EMS#                                           ==00066
==EMS#  SUBJECT:  ACI-TKN-GENERATOR-L           ==00067
==EMS#                                           ==00068
==EMS#  EMPHASIS:  Critical                      ==00069
==EMS#                                           ==00070
==EMS#  ACTION ATTENTION EVENT:  No              ==00071
==EMS#                                           ==00072
==EM2#  PROBABLE CAUSE:                         ==00073
==EMS#    The ANS Object cannot get the space required for object ==00074
==EMS#    private data due to insufficient memory.  The status   ==00075
==EMS#    information returned by DEFINEPOOL indicates the reason ==00076
==EMS#    for the failure of the define operation.              ==00077
==EMS#                                           ==00078
==EM3#  ACTION TAKEN:                           ==00079
==EMS#    The process abends.                      ==00080
==EMS#                                           ==00081
==EM4#  RECOMMENDED ACTION:                     ==00082
==EMS#    Restart the process and send the saveabend file to HELP24.==00083
==EMS#                                           ==00084
==EMS#####==00085
```

The following is an example of the event message documentation output generated by the Manual Generator utility for the above flower box example. This example was generated using the STANDARD option.

NUM: 00010
SEV: Critical (3)
TXT: DEFINEPOOL FAILURE - STATUS = <1>

VAR: <1>: Type: ZSPI-TYP-INT
VA2: Descr: DEFINEPOOL status

CAUSE:

The ANS Object cannot get the space required for object private data due to insufficient memory. The status information returned by DEFINEPOOL indicates the reason for the failure of the define operation.

The process abends.

REMEDY:

Restart the process and send the saveabend file to HELP24.

TAL Library Source File Standards

For the Manual Generator utility to work properly, TAL library documentation must be provided in a specific format within TAL library source files. The TAL library source files should follow ACI file naming conventions. These conventions and format requirements are discussed below.

File Names

The Manual Generator utility imposes no file naming restrictions on TAL library source files, however, the ACI naming convention for TAL source files is `xxxxxxTS`, where `xxxxxx` is a variable number of characters identifying the content of the file. The Manual Generator utility uses the TAL source file name, minus the last two characters of the file name, to create the running header used throughout the generated documentation. For example, if the TAL library source file is named `OKLIBTS`, the running header in the output documentation would be “Oklib Utilities.”

Format Requirements

Within a TAL library source file, all TAL library documentation is derived from comments. All comments in TAL library source files are denoted using the TAL comment characters—two hyphens (`--`). All characters following the two hyphens to the end of the line are considered to be comments.

Individual TAL library documentation is provided in the TAL library source file in the form of commented *flower boxes*. A flower box is a contiguous series of comment lines in the shape of a box. All comment lines in the flower box for a TAL library begins with either the `--LIB#` or `--LIn#` characters, where *n* is number from 1 to 5. Each number denotes specific TAL library information as shown in the following table.

Number	Denotes:
1	The beginning of documentation for a TAL library. This comment line specifies the name of the library procedure.
2	The beginning of the procedure description.
3	The beginning of the syntax for calling the procedure.

Number	Denotes:
4	The beginning of the descriptions for variables used in the procedure call.
5	The beginning of an example call to the library procedure.

Each time the utility finds a `--LI#` comment line in the source file, it inserts a running subhead or a heading in the work file as shown in the following table. All headings are underlined in the output file.

Number	Work File Heading
1	Running subhead containing the procedure name (e.g., LIB_APPL_BIN_SRCH).
2	NARRATIVE
3	PROCEDURE TEMPLATE
4	VARIABLE DESCRIPTION
5	EXAMPLE

When the Manual Generator utility finds the “`--LI1#`” characters, it begins extracting information for a library procedure. It continues to extract information for a procedure until it finds the “`--LIB`” characters followed by five pound sign (#) characters (#####). This signals the end of the documentation for a procedure.

The Manual Generator utility requires that TAL library flower boxes adhere to the BASE24 release 5.0 flower box standards as depicted in the template below. The Manual Generator utility strips out all comment characters, LIB, LI#, and # characters, and all leading and trailing spaces for the generated output.

```
--LIB#####
--LIB#
--LI1#  LIB_PROC_NAME
--LIB#
--LI2#  A description of the procedure goes here. Below follows a
--LIB#  sample call and a description of the variables used in the
--LIB#  call and their data types.
--LIB#
--LI3#  <stat> := LIB_PROC_NAME(
```

```

--LIB#          <param1>,          ! i      req      #
--LIB#          <param2> );          ! o      req      #
--LIB#                                                    #
--LI4#  STAT                                                    #
--LIB#      INT:value                                                    #
--LIB#      Returns one of the values listed below.                    #
--LIB#          LIB_OK_L - The procedure completed successfully.        #
--LIB#          LIB_STAT_DETL_SEE_L - An error occurred. Refer          #
--LIB#              to the STAT_DETL field for further                  #
--LIB#              details.                                            #
--LIB#                                                    #
--LIB#  PARAM1                                                    #
--LIB#      INT .EXT:ref:$len(lib_param1_def)                          #
--LIB#      If STAT is LIB_STAT_DETL_SEE_L, this structure returns      #
--LIB#      detail describing the reason for the failure. The          #
--LIB#      following error values are allowed:                        #
--LIB#                                                    #
--LIB#          LIB_TXN_ERR_L, TXN_STRUCT_LVL_INVLD_L -                #
--LIB#              The structure level for the given context code      #
--LIB#              was invalid.                                         #
--LIB#                                                    #
--LIB#          LIB_TXN_ERR_L, TXN_TCD_CDE_INVLD_L -                   #
--LIB#              An invalid transaction context code was            #
--LIB#              encountered.                                         #
--LIB#                                                    #
--LIB#  PARAM2                                                    #
--LIB#      INT:value                                                    #
--LIB#      The structure or substructure level for which              #
--LIB#      information is required. Supported codes are:              #
--LIB#                                                    #
--LIB#          0 = SUPP_CODE1_L                                         #
--LIB#          1 = SUPP_CODE2_L                                         #
--LIB#                                                    #
--LI5#  struct .param1( lib_param1_def );                              #
--LIB#                                                    #
--LIB#  int      param2;                                                #
--LIB#                                                    #
--LIB#  stat := lib_proc_name( param1, param2 );                        #
--LIB#  if not stat then                                                #
--LIB#      begin                                                        #
--LIB#          lib_ok_l -> ;                                            #
--LIB#          otherwise _.                                              #
--LIB#          ... ! error                                              #
--LIB#      end;                                                        #
--LIB#                                                    #
--LIB######

```

Example TAL Library Source Flower Box

The following is an example of the flower box comments for the LIB_APPL_BIN_SRCH library procedure in the OKLIBTS kernel TAL library source file. The corresponding output generated by the Manual Generator utility for this procedure is provided immediately following this example.

```
--LIB##### !00318
--LIB# # !00319
--LI1# LIB_APPL_BIN_SRCH # !00320
--LIB# # !00321
--LI2# This procedure searches a table for a specified entry, # !00322
--LIB# returning the index and address within the table where a # !00323
--LIB# match was found. The table is assumed to be sorted in # !00324
--LIB# ascending order. The algorithm is from page 58 of # !00325
--LIB# "Algorithms & Data Structures" by Niklaus Wirth. This # !00326
--LIB# routine required less than or equal to log (number of items) # !00327
--LIB# compares to locate any item in an ascendingly sequenced # !00328
--LIB# table. # !00329
--LIB# # !00330
--LIB# Note: When using the entry retrieved, set the structure to # !00331
--LIB# the address retrieved. There is a TAL limitation on the # !00332
--LIB# calculated offset from the structure to the entry index # !00333
--LIB# which traps if the limitation is exceeded. # !00334
--LIB# For example, # !00335
--LIB# Given the declarations: # !00336
--LIB# string .ext data_rec( data_rec_def ) := tbl_addr; # !00337
--LIB# int(32) addr, ! address returned from lib_appl_bin_srch #!!00338
--LIB# idx; ! index returned from lib_appl_bin_srch #!!00339
--LIB# To examine an entry based on an index, assuming the index # !00340
--LIB# is always less than 32767, you could access the record as # !00341
--LIB# 1) data_rec[ $int( idx ) ] or # !00342
--LIB# 2) @data_rec := tbl_addr + # !00343
--LIB# ( $dbl( $len( data_rec ) ) * idx ); or # !00344
--LIB# 3) @data_rec := addr; # !00345
--LIB# Example 1 could result in a programmatic trap if the offset # !00346
--LIB# exceeds the TAL limitation or be invalid if the number of # !00347
--LIB# entries ever exceeds 32767. # !00348
--LIB# Example 2 is acceptable. # !00349
--LIB# Example 3 is the preferred method. # !00350
--LIB# # !00351
--LI3# <stat> := LIB_APPL_BIN_SRCH( # !00352
--LIB# <stat_det1>, ! o req #!!00353
--LIB# <tbl>, ! i req #!!00354
--LIB# <tbl_rec_size>, ! i req #!!00355
--LIB# <tbl_num_recs>, ! i req #!!00356
--LIB# <key_ofst>, ! i req #!!00357
--LIB# <key_lgth>, ! i req #!!00358
```

```

--LIB#          <srch_tkn>,          ! i      req      #!!00359
--LIB#          <idx>,                ! o      opt      #!!00360
--LIB#          <addr> );              ! o      opt      #!!00361
--LIB#                                                  # !00362
--LI4#  STAT                                                  # !00363
--LIB#          INT:value              returned value      # !00364
--LIB#          This value indicates the outcome of the call. Possible # !00365
--LIB#          values are listed below:                      # !00366
--LIB#          LIB_OK_L                - successful call    # !00367
--LIB#          LIB_NOT_FOUND_L         - table entry not found # !00368
--LIB#          LIB_PARAM_MISS_L       - missing required parameter # !00369
--LIB#          LIB_STAT_DETL_SEE_L    - error              # !00370
--LIB#                                                  # !00371
--LIB#  STAT_DETL                                              # !00372
--LIB#          INT .EXT:ref:$len(lib_stat_detl_def)        # !00373
--LIB#          If STAT is LIB_STAT_DETL_SEE_L, this structure returns # !00374
--LIB#          detail describing the reason for the failure.  # !00375
--LIB#                                                  # !00376
--LIB#          LIB_ERR_L, LIB_TBL_REC_NUM_L                 # !00377
--LIB#          The number of records must be greater than zero. # !00378
--LIB#                                                  # !00379
--LIB#  TBL                                                    # !00380
--LIB#          STRING .EXT:ref                                # !00381
--LIB#          The table to be searched.                     # !00382
--LIB#                                                  # !00383
--LIB#  TBL_REC_SIZE                                           # !00384
--LIB#          INT                                            # !00385
--LIB#          The length of each table entry in bytes.      # !00386
--LIB#                                                  # !00387
--LIB#  TBL_NUM_RECS                                           # !00388
--LIB#          INT(32)                                        # !00389
--LIB#          The number of entries in the table. Maximum 2,147,483,647 # !00390
--LIB#                                                  # !00391
--LIB#  KEY_OFST                                               # !00392
--LIB#          INT                                            # !00393
--LIB#          The byte offset of the key field within a table entry. # !00394
--LIB#                                                  # !00395
--LIB#  KEY_LGTH                                               # !00396
--LIB#          INT                                            # !00397
--LIB#          The length of the key field in bytes.         # !00398
--LIB#                                                  # !00399
--LIB#  SRCH_TKN                                              # !00400
--LIB#          STRING .EXT:ref                                # !00401
--LIB#          The search token to match on as the key in the table. # !00402
--LIB#                                                  # !00403
--LIB#  IDX                                                    # !00404
--LIB#          INT(32) .EXT:ref                                # !00405
--LIB#          The table index of the entry that matches the search # !00406
--LIB#          token, relative to zero.                      # !00407
--LIB#                                                  # !00408
--LIB#  ADDR                                                  # !00409
--LIB#          INT(32) .EXT:ref                                # !00410

```

```

--LIB#      The byte address of the entry that matches the search      # !00411
--LIB#      token in the table.                                         # !00412
--LIB#                                             # !00413
--LI5#  struct .stat_det1( lib_stat_det1_def );                         # !00414
--LIB#                                             # !00415
--LIB#  string .ext tbl,                                                # !00416
--LIB#          .ext key;                                                # !00417
--LIB#  int      rec_siz,                                                # !00418
--LIB#          key_ofst,                                                # !00419
--LIB#          key_lgth,                                                # !00420
--LIB#          stat;                                                    # !00421
--LIB#  int(32) addr,                                                    # !00422
--LIB#          idx,                                                     # !00423
--LIB#          num_recs;                                                # !00424
--LIB#                                             # !00425
--LIB#  stat := lib_appl_bin_srch( stat_det1, tbl, rec_siz,             # !00426
--LIB#                                     num_recs, key_ofst, key_lgth,     # !00427
--LIB#                                     key, idx, addr );                 # !00428
--LIB#  case stat of                                                    # !00429
--LIB#      begin                                                         # !00430
--LIB#          lib_ok_1 -> ;                                              # !00431
--LIB#          otherwise ->                                              # !00432
--LIB#          ... ! error                                              #!!00433
--LIB#          end;                                                      # !00434
--LIB#                                             # !00435
--LIB##### !00436

```

The following is an example of the documentation generated by the Manual Generator utility for the above flower box example. This example was generated using the FORMATTED option.

```

\head "Oklib  Utilities          "
\subhead "LIB_APPL_BIN_SRCH      "
\new
\need B
\ov 2
NARRATIVE
NARRATIVE

```

This procedure searches a table for a specified entry, returning the index and address within the table where a match was found. The table is assumed to be sorted in ascending order. The algorithm is from page 58 of "Algorithms & Data Structures" by Niklaus Wirth. This routine required less than or equal to log (number of items) compares to locate any item in an ascendingly sequenced table.

Note: When using the entry retrieved, set the structure to the address retrieved. There is a TAL limitation on the calculated offset from the structure to the entry index which traps if the limitation is exceeded.

For example,

Given the declarations:

```
string .ext data_rec( data_rec_def ) := tbl_addr;
int(32) addr, ! address returned from lib_appl_bin_srch
           idx; ! index returned from lib_appl_bin_srch
```

To examine an entry based on an index, assuming the index is always less than 32767, you could access the record as

```
1) data_rec[ $int( idx ) ]           or
2) @data_rec := tbl_addr +
   ( $dbl( $len( data_rec ) ) * idx ); or
3) @data_rec := addr;
```

Example 1 could result in a programmatic trap if the offset exceeds the TAL limitation or be invalid if the number of entries ever exceeds 32767.

Example 2 is acceptable.

Example 3 is the preferred method.

\need B

\ov 2

PROCEDURE TEMPLATE

PROCEDURE TEMPLATE

```
<stat> := LIB_APPL_BIN_SRCH(
    <stat_detl>,           ! o    req
    <tbl>,                 ! i    req
    <tbl_rec_size>,        ! i    req
    <tbl_num_recs>,        ! i    req
    <key_ofst>,            ! i    req
    <key_lgth>,            ! i    req
    <srch_tkn>,            ! i    req
    <idx>,                 ! o    opt
    <addr> );              ! o    opt
```

\need B

\ov 2

VARIABLE DESCRIPTION

VARIABLE DESCRIPTION

STAT

INT:value returned value

This value indicates the outcome of the call. Possible values are listed below:

```
LIB_OK_L           - successful call
LIB_NOT_FOUND_L    - table entry not found
LIB_PARAM_MISS_L   - missing required parameter
LIB_STAT_DETL_SEE_L - error
```

STAT_DETL

INT .EXT:ref:\$len(lib_stat_detl_def)

If STAT is LIB_STAT_DETL_SEE_L, this structure returns detail describing the reason for the failure.

LIB_ERR_L, LIB_TBL_REC_NUM_L

The number of records must be greater than zero.

TBL

STRING .EXT:ref

The table to be searched.

TBL_REC_SIZE

INT

The length of each table entry in bytes.

TBL_NUM_RECS

INT(32)

The number of entries in the table. Maximum 2,147,483,647

KEY_OFST

INT

The byte offset of the key field within a table entry.

KEY_LGTH

INT

The length of the key field in bytes.

SRCH_TKN

STRING .EXT:ref

The search token to match on as the key in the table.

IDX

INT(32) .EXT:ref

The table index of the entry that matches the search token, relative to zero.

ADDR

INT(32) .EXT:ref

The byte address of the entry that matches the search token in the table.

\need B

\ov 2

EXAMPLE

EXAMPLE

```
struct .stat_detl( lib_stat_detl_def );
```

```
string .ext tbl,
```

```
        .ext key;
```

```
int     rec_siz,
```

```
        key_ofst,
        key_lgth,
        stat;
int(32) addr,
        idx,
        num_recs;

stat := lib_appl_bin_srch( stat_det1, tbl, rec_siz,
                          num_recs, key_ofst, key_lgth,
                          key, idx, addr );

case stat of
    begin
        lib_ok_1 -> ;
    otherwise ->
        ... ! error
    end;
```


Title Pages

The Manual Generator utility allows you to attach a title page to documentation files for printing from a spooler. If you specify a title page when the Manual Generator utility is run, the following command is added to the beginning of the output file:

```
\in $vol.subvol.title-page-file-name
```

This command sources in the specified title page file when the output file is printed using TGAL. ACI provides a default title page file named MGENTITL with the Manual Generator utility. This file contains an ACI logo and copyright information. You can create your own customized title page file if desired.

The title page file name is specified at a prompt when the Manual Generator utility is run interactively or using the TITLEPAGE-FILE assign when the utility is run noninteractively. The TITLE option must also be set when running the utility noninteractively.

The following TGAL commands and text is added by the Manual Generator utility to the bottom of the title page. This text identifies the source file used to generate the documentation and the TGAL command syntax used to print the documentation.

```
\center on
      Input File: $data.ok60ans.ansps

      TGAL command to recreate this manual:

      TGAL /IN $data.ok60logm.isknans, OUT $S.#ans/
\center off
```

Table of Contents

When requested, the Manual Generator utility automatically generates a table of contents for TAL library procedure documentation. This table of contents contains the name of each library procedure included and the page number on which each procedure begins. The utility uses the input file name minus the last two characters in a running head for the table of contents. Page numbers in the table of contents are prefaced with the “TOC” characters.

The following is a partial example table of contents output generated by the Manual Generator utility for core TAL library procedures. The source file for this example is OC60LIB.OCLIBTS.

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LIB_DATACONV_ACC_NUM	LIB-5
LIB_DATTIM_PRD_BEG	LIB-7
LIB_DATTIM_PRD_BEG_DAYS	LIB-10
LIB_DATTIM_PRD_BEG_LGTH	LIB-12
LIB_NAM_TOKENIZE_SOUNDEX	LIB-14

LAST CHANGE DATE: MM/DD/YY

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